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Your Signature (do not print):

Liu Rui

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LIU RUI

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Form 62D**The ACT[®]****2008 | 2009**

Directions

This booklet contains tests in English, Mathematics, Reading, and Science. These tests measure skills and abilities highly related to high school course work and success in college. **CALCULATORS MAY BE USED ON THE MATHEMATICS TEST ONLY.**

The questions in each test are numbered, and the suggested answers for each question are lettered. On the answer document, the rows of ovals are numbered to match the questions, and the ovals in each row are lettered to correspond to the suggested answers.

For each question, first decide which answer is best. Next, locate on the answer document the row of ovals numbered the same as the question. Then, locate the oval in that row lettered the same as your answer. Finally, fill in the oval completely. Use a soft lead pencil and make your marks heavy and black. **DO NOT USE A BALLPOINT PEN.**

Mark only one answer to each question. If you change your mind about an answer, erase your first mark thoroughly before marking your new answer. For each question, make certain that you mark in the row of ovals with the same number as the question.

Only responses marked on your answer document will be scored. Your score on each test will be based only on the number of questions you answer correctly during the time allowed for that test. You will NOT be penalized for guessing. **IT IS TO YOUR ADVANTAGE TO ANSWER EVERY QUESTION EVEN IF YOU MUST GUESS.**

You may work on each test ONLY when your test supervisor tells you to do so. If you finish a test before time is called for that test, you should use the time remaining to reconsider questions you are uncertain about in that test. You may NOT look back to a test on which time has already been called, and you may NOT go ahead to another test. To do so will disqualify you from the examination.

Lay your pencil down immediately when time is called at the end of each test. You may NOT for any reason fill in or alter ovals for a test after time is called for that test. To do so will disqualify you from the examination.

Do not fold or tear the pages of your test booklet.

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ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE." In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

The Book That Helped Save the Everglades

[1]

Few books have been

1

more influential in the U.S. environmental
movement as Marjory Stoneman Douglas's

The Everglades: River of Grass. Published in 1947,
the book educated the nation about the dangers facing
one of the most fragile ecosystems and said its author's
emergence as a famous conservation crusader.

[2]

Douglas had moved to Florida when she was
twenty-five, working first to a newspaper reporter

4

1. Which of the following alternatives to the underlined portion would NOT be acceptable?

A. Scarcely any
B. Hardly any
C. Not much
D. Not many

2. F. NO CHANGE

G. as
H. as if
J. OMIT the underlined portion.

3. A. NO CHANGE

B. marked
C. remarked
D. made

4. F. NO CHANGE

G. as
H. in
J. at

and then as a freelance writer. ⁵ In 1942 an editor

who also wrote novels asked her to write an in-depth report about the distinct features of the Miami River, a topic she broadened to include the Everglades.

For five years she participated in a study of the great wetlands that once covered millions of acres in southern Florida, learning about their complex biology, geology, and history. ⁸

[3]

The more Douglas studied the region, the more she became inspired by the beauty to be found in nature. Though a portion of the ecosystem had been preserved in 1934 as Everglades National Park, the remainder faced relentless human pressures. She watched in growing

dismay. As swampy areas were drained and tracts of houses replaced stands of saw grass.

5. At this point, the writer is considering adding the following true statement:

Her first exposure to the state was at the age of four, when she joined her parents on a business trip to Tampa.

Would this be a relevant addition to make here?

- A. Yes, because it provides interesting background information about Douglas's life.
B. Yes, because it puts Douglas's move to Florida into perspective for the reader.
C. No, because not enough detail is provided about the duration of that first visit.
D. No, because it interrupts the sequence of events marking her career development.
6. F. NO CHANGE
G. who also was a novelist
H. that also wrote novels
J. OMIT the underlined portion.
7. Which choice best expresses Douglas's intense involvement in her study of the wetlands?
A. NO CHANGE
B. immersed herself in
C. gave thought to
D. dabbled in
8. The writer is considering deleting the following phrase from the preceding sentence (replacing the comma after *Florida* with a period):
learning about their complex biology, geology, and history.
Should the writer make this deletion?
F. Yes, because the phrase limits the extent of Douglas's overall study.
G. Yes, because the relationship between the phrase and Douglas's investigation is not made evident to the reader.
H. No, because the phrase provides revealing background information about Douglas herself.
J. No, because the phrase identifies the scope of Douglas's investigation.
9. Which choice most effectively completes the sentence and introduces the subject of Paragraph 3?
A. NO CHANGE
B. alarmed by what was happening to it.
C. fascinated by the distinct features of the region.
D. intrigued by its various aspects.
10. F. NO CHANGE
G. relentlessly humanly
H. human relentlessly
J. relentless and humanly
11. A. NO CHANGE
B. dismay; as
C. dismay as
D. dismay as,



[4]

In publishing *The Everglades*, Douglas offered an eloquent rallying cry to defend a little-understood ecosystem. Due to the publicity of attracting public interest in her book, Douglas¹² was able to take her message directly to the people of Florida, giving countless speeches about the Everglades' importance.

[5]

Thanks in part to Douglas's book, the Everglades are now regarded as a national treasure. Though still threatened, they form a very fragile ecosystem. In¹³ recognition of her work, Douglas later received the

nation's civilian highest honor, the presidential Medal¹⁴ of Freedom, and many conservation awards. Her most significant legacy, however, remains the intricate web of water, grass, and wildlife that her book helped to protect.

12. F. NO CHANGE
G. created by her book,
H. of her book receiving public acclaim,
J. resulting from the public attention to her book,

13. Given that all the choices are true, which one most clearly suggests the impact Douglas's work may have had on the Everglades?
A. NO CHANGE
B. were long the subject of Douglas's devotion and concern.
C. are biologically, geologically, and historically complex.
D. are far healthier than when the book first appeared.

14. F. NO CHANGE
G. nation's highest civilian
H. civilian nation's highest
J. highest nation's civilian

Question 15 asks about the preceding passage as a whole.

15. Upon reviewing the essay and finding that some information has been left out, the writer composes the following sentence incorporating that information:

Increasing numbers of birds and fish were dying each year as a result of this interruption of the natural cycles upon which the wetland system depended.

This sentence would most logically be placed after the last sentence in Paragraph:

- A. 2.
B. 3.
C. 4.
D. 5.

PASSAGE II

Thinking about Pennies

There's a movement underway to get rid of the penny.

Proponents of its abolition argue that the cost of minting

16

pennies and keeping them in circulation far outstrips

17

their collective worth. The thinking about pennies are

18

a nuisance. What they ask can anyone buy with a

19

penny (or two or three) these days? Not much, right?

[1] It pays to remember, however, that the modern

penny has history on its side. [2] It is the descendant of

our country's first initial federally authorized coin, the

20

Fugio Cent. [3] This coin, issued in 1787, came complete

with free advice. [4] Below this was the admonishment to

"mind your business." [5] The word *fugio* ("I fly" in Latin)

was stamped on one side. 21

Nowadays it may be hard to imagine that a mere

22

penny was ever worth minding (let alone saving), but there

was a time when even a piece of a penny was something

you could trade. Until the reign of Edward I (1272–1307),

the back of the British penny was stamped with a cross so

that they could easily be cut into halves (a half-penny,

23

pronounced in England as "hay-penny") and are even

24

into fourths (a farthing).

16. F. NO CHANGE

G. its

H. their

J. his

17. A. NO CHANGE

B. would have far outstripped

C. was far outstripping

D. will be far outstripping

18. F. NO CHANGE

G. Thinking

H. They think

J. The thought of

19. A. NO CHANGE

B. What, they ask

C. What they ask,

D. What, they ask,

20. F. NO CHANGE

G. earliest initial

H. very first

J. primary original

21. For the sake of the logic and coherence of this paragraph, Sentence 4 should be placed:

A. where it is now.

B. after Sentence 1.

C. after Sentence 2.

D. after Sentence 5.

22. Which choice provides the most effective beginning to this paragraph?

F. NO CHANGE

G. Few would have believed that these days

H. Just as it was back then, these days

J. Isn't it interesting that these days

23. A. NO CHANGE

B. it

C. ours

D. he

24. F. NO CHANGE

G. are divided in four

H. are quartered

J. even

Of course, these days even an entire untouched penny

hardly lying on the sidewalk is worth bending over to

25

pick up. Maybe it's because pennies are, everywhere the

26

U.S. Mint report's that it produced 10,257,400,000 pennies

27

in 1998. That's an average of over 28 million pennies per day!

Oddly enough, though, the fact that there existence is

28

in such huge numbers and are seen as "worthless" is what allows pennies to enrich our lives in a small but priceless way. Consider how rarely any of us arrives at a checkout counter with exact change. Since it is at precisely this

29

moment that the true value of the penny reveals itself.

When a clerk dips into that little dish by the register for an extra penny or two, or when a customer replenishes that same dish it allows for the kind of simple act of

30

courtesy found less and less often in modern life. And you still think the penny is worthless?

25. The best placement for the underlined portion would be:

- A. where it is now.
- B. after the word *days*.
- C. after the word *is*.
- D. after the word *worth*.

26. F. NO CHANGE
G. are everywhere:
H. are everywhere,
J. are everywhere

27. A. NO CHANGE
B. Mint's reports
C. Mint reports
D. Mint reports'

28. F. NO CHANGE
G. their existence is
H. although they exist
J. they exist

29. A. NO CHANGE
B. On the other hand, it
C. However, it
D. It

30. F. NO CHANGE
G. dish it,
H. dish, it
J. dish; it

PASSAGE III

Nut Harvesting in the Amazon

Brazil nut trees are found not only in Brazil. One of the most productive areas for gathering the tasty, nutritious Brazil nut is in a remote corner of Peru in the Amazon Rain Forest. Enrique Ortiz, a biologist for the National Museum of Natural History, knows the area well. For more than eight years, Ortiz has been studying Brazil nuts.

1 ■ ■ ■ ■ ■ ■ ■ ■ 1

A single pod of Brazil nuts (which are actually seeds) weighs 31 nearly six pounds, so when it falls from a tree eight stories high, it has the 31 force to cause serious injury.

Despite the danger, Ortiz and his fellow scientists are committed to their research 32 project, the team of eight keeps watch over about 1,000 Brazil nut trees. They count every pod that falls and 33 tally them so that *castañeros* (nut harvesters) can gather the most nuts in the least amount of time.

As Ortiz well knows, the *castañeros* must time 34 their nut gathering perfectly. The mature pods, each holding up to twenty-five Brazil nuts, fall during Peru's rainy season. If the *castañeros* arrive too early, they must wait for the pods to fall. If they arrive too late, large 35 rodents called agoutis will

www.crackab.com

have buried the pods to 36 eat later

for future meals. 37

31. A. NO CHANGE
B. weigh nearly
C. weighing nearly
D. weigh near
32. F. NO CHANGE
G. project, the team of eight,
H. project: the team of eight,
J. project: the team of eight
33. Given that all the choices are true, which one would draw the clearest connection between the work of research scientists and the work of the nut harvesters?
A. NO CHANGE
B. admire the nut trees' blossoms
C. mark the most productive areas
D. are diligent in their work
34. Given that all the choices are true, which one most effectively leads the reader from the preceding paragraph to the subject of this new paragraph?
F. NO CHANGE
G. As he continues his work in Peru, Ortiz has several future goals in mind.
H. After the *castañeros* gather their nuts, the nuts are picked up and taken to market.
J. A sack of Brazil nuts weighs up to 140 pounds.
35. A. NO CHANGE
B. rodents, called agoutis
C. rodents called agoutis,
D. rodents called, agoutis
36. F. NO CHANGE
G. to have subsequently
H. to consume at a later point
J. OMIT the underlined portion.
37. At this point, the writer is considering adding the following true statement:
While gathering nuts, the *castañeros* come in contact with vipers and jaguars, and they sometimes contract malaria.
Should the writer make this addition here?
A. Yes, because it further supports the main idea of this paragraph.
B. Yes, because it provides an essential link to the paragraph that follows.
C. No, because it distracts the reader from the paragraph's focus at this point on the agoutis.
D. No, because it implies that the work of the *castañeros* is dangerous, which contradicts information presented earlier in the essay.

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

Ortiz has also been studying how the agouti helps the Brazil nut tree to survive. The agouti's strong teeth can open the thick pods to free the seeds that may eventually

sprout into future trees. Complicating their goals, Ortiz and his team opened 120 pods, attached a magnetic strip

to each seed, and the pods were glued back together. The agoutis rounded up the pods as usual. Six months later,

the researchers can be able to track down the sprouted seeds to find out how far the rodents had carried them.

Ortiz's painstaking work provides a clear picture of how the Brazil nut tree, with the help of the agouti,

replenish themselves. The survival of the tree is essential

to the South American economy, since about annually \$44 million worth of Brazil nuts are sold.

38. F. NO CHANGE
G. spending moments in reflection trying to think of the ways the agouti can promote the survival of the Brazil nut tree.
H. turning his attention toward the survival of the Brazil nut tree in terms of the agouti.
J. doing his research in the area of the agouti, who has a meaningful part in the way in which the Brazil nut tree will survive.
39. Which choice provides the best transition from the preceding sentence to this one?
A. NO CHANGE
B. Toward a more meaningful understanding,
C. As it happened,
D. To understand better the agouti's role,
40. F. NO CHANGE
G. gluing the pods
H. glue the pods
J. glued the pods
41. A. NO CHANGE
B. have been able
C. were
D. could be
42. Which of the following alternatives to the underlined portion would be LEAST acceptable?
F. diligent
G. rigid
H. careful
J. thorough
43. A. NO CHANGE
B. replenishes itself.
C. replenish itself.
D. replenishes themselves.
44. The best placement for the underlined portion would be:
F. where it is now.
G. after the word *million*.
H. after the word *worth*.
J. after the word *sold* (ending the sentence with a period).

Question 45 asks about the preceding passage as a whole.

45. Suppose the writer had intended to write a brief essay focusing on an animal that lives in the South American rain forest. Would this essay successfully fulfill the writer's goal?
- A. No, because the essay primarily focuses on scientific research involving the Brazil nut tree.
 - B. No, because the essay describes the agouti but does not indicate whether it lives in the South American rain forest.
 - C. Yes, because the essay focuses on the eating habits of the agouti in the South American rain forest.
 - D. Yes, because the essay emphasizes one scientist's commitment to saving rain forest creatures.

PASSAGE IV

A Beach in Brooklyn

On the Fourth of July, my friend Jeremy and I rode the subway to the last stop: Coney Island. Jeremy's grandparents had loved to go there many years ago and when we wanted to see what the place was like today. As soon as we stepped off the subway into the sweltering, salty air—air so bright it glinted like chrome—and seeing the ancient amusement park rides looming in the distance, tall as water towers.

On the boardwalk a woman is selling peeled mangoes for a dollar each. She served them doused with lemon and salt, and cut so expertly that plucking off the slices was easy. They tasted zingy and refreshing.

46. F. NO CHANGE
G. ago and that
H. ago, so
J. ago,
47. A. NO CHANGE
B. we could see
C. which seeing
D. then seeing
48. If the writer were to delete the words *sweltering*, *salty*, and *ancient* from the preceding sentence, the paragraph would primarily lose:
- F. elements of the setting.
 - G. details that reveal character traits.
 - H. details that add mystery.
 - J. a crucial link to the next paragraph.
49. A. NO CHANGE
B. sells
C. will be selling
D. sold
50. F. NO CHANGE
G. expertly as
H. expert that
J. expert as

Jeremy wanted to ride the Cyclone—the famous roller coaster that plunges and loops, sketching infinity signs

51

across the sky. “Please go alone,” I said, watching the riders careen past, lifting their arms and shrieking in fear and delight.

52

Above all, we boarded the Wonder

53

Wheel, a Ferris wheel, that looked as tall as a

54

seven-story building. We ascended slowly upward, with the operator stopping the ride often to let people board. For a long time our caged bucket dangled at the top, rocking, as we took in the magnificent view: the Atlantic Ocean flinging itself onto the shore, the beach and surf dotted with people the size of letters in a newspaper. At last everyone was on board. The huge wheel rotated once, and then we were back on the ground and the door to our bucket opened. Jeremy laughed. Such a short ride! Still, for that splendid view, it was worth it.

55

56

57

[1] The place has certainly gotten smaller since those days. [2] The amusement area has shrunk from twenty blocks to just three, but it’s still glorious, still perfect for a July expedition. [3] On the subway going home, I remembered photos I’d seen of Coney Island in the early 1900s—black-and-white images of strongmen in striped shirts, women were under parasols, and rides that looked

58

59

51. A. NO CHANGE
B. loops; sketching
C. loops. Sketching
D. loops sketching
52. Which of the following alternatives to the underlined portion would NOT be acceptable?
F. said when I watched
G. said as I watched
H. said, while watching
J. said. Watching
53. A. NO CHANGE
B. With appreciation
C. For instance,
D. Instead.
54. F. NO CHANGE
G. Wheel, a Ferris wheel
H. Wheel a Ferris wheel,
J. Wheel; a Ferris wheel
55. A. NO CHANGE
B. in upward motion, slowly
C. slowly,
D. upward slowly,
56. F. NO CHANGE
G. view;
H. view
J. view, of
57. A. NO CHANGE
B. concluded a singular solitary rotation,
C. revolved in a single rotation,
D. made one rotation all the way around,
58. F. NO CHANGE
G. shrinked
H. has shrank
J. had shrank
59. A. NO CHANGE
B. under parasols, and rides that
C. are seen under parasols, and rides
D. under parasols, and rides



rusty and treacherous even then. 60

60. Which of the following sequences of sentences make this paragraph most logical?
- F. NO CHANGE
 - G. 2, 1, 3
 - H. 3, 2, 1
 - J. 3, 1, 2

PASSAGE V

An Endangered Amusement

While on vacation recently, I was surprised to see an antique carousel in a modern shopping mall. 61 Though it had been years since I had ridden one of these merry-go-rounds, I instinctively did what I would of done 62 as a girl: I searched for "my" horse. This time, it is a 63 handsome cream-colored steed on the outside row, with brightly painted gear that glittered with jewels. Although the gems were fake, they caught my eye, just as the designers had planned. 64

61. If the writer deleted the phrase "in a modern shopping mall" from the preceding sentence (placing a period after the word *carousel*), the essay would primarily lose a detail that:
- A. explains where the narrator was looking for an antique carousel.
 - B. helps identify the setting that accounts in part for the narrator's surprise.
 - C. refers to the setting, which is described in detail later in the essay.
 - D. establishes where the narrator was surprised to be during a vacation.
62. F. NO CHANGE
G. might of
H. had
J. OMIT the underlined portion.
63. A. NO CHANGE
B. will be
C. had been
D. was
64. F. NO CHANGE
G. planned, as a way to make the carousel eye-appealing to onlookers.
H. planned in their intentions for the carousel I was riding.
J. planned and intended as they designed the carousel.



[1] Once in the saddle, I admired the other details of my horse: the carved wooden mane, the bared teeth, and the tail made of real horsehair. [2] When I reached down for the reins, I noticed the most stunning feature of my

horse: its eyes. [3] Eerily real, they gave me the impression that somebody inhabited that wooden frame and was

looking back at me. [4] As I waited for the ride to begin, children appeared from nowhere. [5] With excitement in their voices, they chose their mounts,

a challenging process for them, or so it seemed.

[6] At last, the bell clanged, the lights came on, and the music began. [7] The animals moved

smoothly up and down as the carousel began to turn. [8] I glanced to my left and noticed a little girl next to me, probably about seven years old.

[9] Her eyes sparkled, and, fists on her hips, she announced to no one in particular, "Look!

No hands!"

65. Given that all of the choices are true, which one best provides information about the carousel that is specific and new?

- A. NO CHANGE
- B. the features that gave me the sense that this was an antique carousel, not a modern one.
- C. the jewels, the cream-colored paint, the eyes that caught my attention.
- D. the mane, the teeth, the tail, and the cream-colored paint.

66. F. NO CHANGE

- G. it's
- H. its'
- J. both of their

67. Which of the following alternatives to the underlined portion would NOT be acceptable?

- A. peering out
- B. glancing back
- C. looking away
- D. looking

68. Which choice best allows the reader to visualize the children in the process of choosing their carousel horses?

- F. NO CHANGE
- G. a process that seemed to take them about the same time as it took me.
- H. sometimes trying a few different animals before making their final choice.
- J. and I wondered if they felt as sure about their choices as I felt about mine.

69. A. NO CHANGE

- B. smoothly up, and down
- C. smoothly up; and down
- D. smoothly up and down;

70. F. NO CHANGE

- G. Her eyes sparkled, fists on her hips, and announced
- H. Fists on her hips, her eyes sparkled, she announced
- J. With the announcement, her eyes sparkled and, fists on her hips,

71. The writer wants to divide the preceding paragraph into two so that one of the paragraphs focuses on the narrator's awareness of the children riding the carousel. The writer can best accomplish this goal by beginning a new paragraph with Sentence:

- A. 3.
- B. 4.
- C. 5.
- D. 6.



He had the answers to some of my questions, and we
⁷²
had a pleasant talk. I strolled over to talk to the operator,
⁷²

he explained that this carousel had been designed,
⁷³
carved, and built in 1906 by Charles Looff. I read on a
plaque nearby that over 5,000 carousels had been built
by artisans, including Looff, between 1880 and 1939
⁷⁴
the golden age of carousels. Now there are fewer than
175 hand-carved wooden carousels still operating in the
United States. Many of them like this one, feature not
⁷⁵
just horses but also lions, giraffes, zebras, and elephants.
They all deserve to be kept running forever.

72. Given that all of the choices are true, which one provides the most effective transition in the narrative from the preceding paragraph to this paragraph?
- F. NO CHANGE
 - G. Later, after I had left the mall, I wished I had taken a picture of the carousel.
 - H. Three minutes later, the spinning platform came to a stop and so did the music.
 - J. Spinning on this carousel, I began to remember the carousels I had ridden during my childhood.
73. A. NO CHANGE
B. and his explanation
C. who explained
D. there he explained
74. F. NO CHANGE
G. artisans including Looff, between 1880 and 1939
H. artisans, including Looff between 1880 and 1939,
J. artisans, including Looff, between 1880 and 1939,
75. A. NO CHANGE
B. them, like this one feature
C. them like this one, feature.
D. them, like this one, feature

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. The polynomial $x^2 - 16$ factors as:

- A. $(x + 4)(x + 4)$
- B. $(x - 4)(x - 4)$
- C. $(x - 4)(x + 4)$
- D. $(x - 16)(x - 16)$
- E. $(x - 16)(x + 16)$

2. At a certain store, 3 bottles of water and 2 bags of chips cost \$8.00. At the same store, 2 identical bottles of water and 3 identical bags of chips cost \$7.00. How much would 2 of these bottles of water and 2 of these bags of chips cost at this store?

- F. \$6.00
- G. \$3.00
- H. \$2.00
- J. \$1.50
- K. \$1.00

3. Tom has scores of 0, 6, 7, 8, 10, 10, and 10 on his science lab reports, which are all equally weighted. Which of the following is closest to the mean of his science lab report scores?

- A. 5
- B. 7.3
- C. 8
- D. 8.5
- E. 10

4. What is the value of $3k^2 - k + 2$ when $k = -3$?

- F. -28
- G. 8
- H. 32
- J. 80
- K. 86

5. The expression $2y - 5 + 3y$ is equivalent to which of the following expressions?

- A. 0
- B. $-y - 5$
- C. $y - 5$
- D. $5y - 5$
- E. $6y - 5$

DO YOUR FIGURING HERE.



6. What is the value of $f(3)$ when $f(x) = 2x^2 + x + 3$?

F. 12
G. 18
H. 21
J. 24
K. 30

DO YOUR FIGURING HERE.

7. The filling machine in a cereal factory fills 1,200 boxes of cereal every 5 minutes. Which of the following proportions can be solved for t , the number of minutes it takes the machine to fill 2,000 of these boxes of cereal?

A. $\frac{1,200}{5} = \frac{2,000}{t}$
B. $\frac{1,200}{2,000} = \frac{t}{5}$
C. $\frac{1,200}{t} = \frac{2,000}{5}$
D. $\frac{1,200}{5} = \frac{2,000 - 1,200}{t}$
E. $\frac{1,200}{5} = \frac{1,200 + 2,000}{t}$

8. If a farmer originally had a supply of 1,450 bushels of corn and sold 580 bushels, what percent of her original supply did she sell?

F. 40%
G. 58%
H. 60%
J. 67%
K. 87%

9. Pierre is sewing a clown outfit for a customer and needs a button. In his button bag, he has 8 yellow buttons, 4 orange buttons, and 6 red buttons, all the same size. Pierre will take one of these buttons out of the bag at random. What is the probability that the button is an orange button?

A. $\frac{1}{18}$
B. $\frac{1}{4}$
C. $\frac{2}{9}$
D. $\frac{2}{7}$
E. $\frac{2}{5}$



10. Given $x^3 = 38.7$, where x is a real number, the value of x is between which 2 consecutive integers?

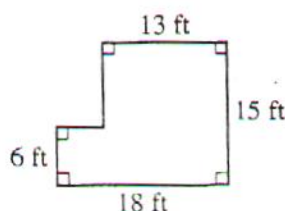
F. 3 and 4
 G. 6 and 7
 H. 12 and 13
 J. 38 and 39
 K. 57,960 and 57,961

DO YOUR FIGURING HERE.

11. When the number 0.00735 is expressed in scientific notation, it is represented by 7.35×10^x . What is the value of x ?

A. -5
 B. -3
 C. -2
 D. 2
 E. 3

12. Terri plans to tile her kitchen floor and needs to know the area of the floor to get an estimate of how much the project will cost. The figure below shows her kitchen floor with dimensions given in feet. What is the area of Terri's kitchen floor, in square feet?



F. 132
 G. 225
 H. 245
 J. 270
 K. 303

13. What is the midpoint of the line segment with endpoints $(-2, 8)$ and $(6, 2)$ in the standard (x, y) coordinate plane?

A. $(-4, 3)$
 B. $(2, 5)$
 C. $(4, -6)$
 D. $(4, 10)$
 E. $(8, -6)$

14. A government class has an enrollment of 26 students, all of whom are juniors or seniors. There are 4 more juniors enrolled than seniors. How many seniors are enrolled in the class?

F. 9
 G. 11
 H. 15
 J. 17
 K. 22



15. In basketball, when a player makes a basket, his or her team scores 3 points if the ball is shot from a distance of more than 6 meters from the basket, 1 point for a free throw, and 2 points for all other types of shots. During a recent game, Will's team scored a total of 26 points on Will's baskets, which consisted of 4 free throws and more 3-point baskets than 2-point baskets. How many 3-point baskets did Will make during the game?

A. 3
B. 4
C. 5
D. 6
E. 7

DO YOUR FIGURING HERE.

16. What real value of x satisfies the equation

$$\frac{x}{200} + \frac{1}{5} = \frac{x+100}{300} ?$$

F. 16
G. 64
H. 80
J. 160
K. 320

17. The time that a storm will take to reach a town can be predicted by using the formula $t = \frac{d}{s}$, where t is the time, in hours; d is the distance, in miles; and s is the speed, in miles per hour. A storm was 200 miles from the edge of Burlington at 10:00 A.M. and was traveling directly toward Burlington at a speed of 25 mph. According to the formula, at what time will the storm reach Burlington if the storm does not change course or speed?

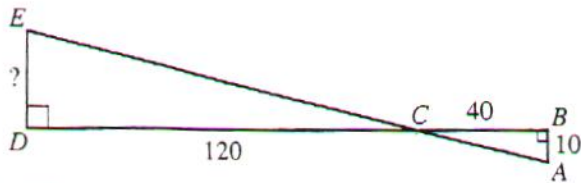
A. 8:00 P.M.
B. 6:00 P.M.
C. 4:00 P.M.
D. 2:00 P.M.
E. 11:00 A.M.

18. A line segment and a plane can have at most how many points in common?

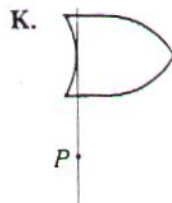
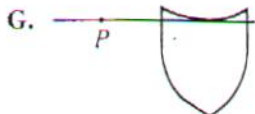
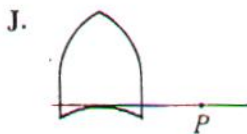
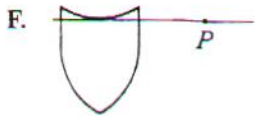
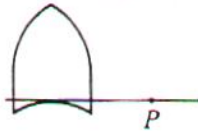
F. 0
G. 1
H. 2
J. 3
K. Infinitely many



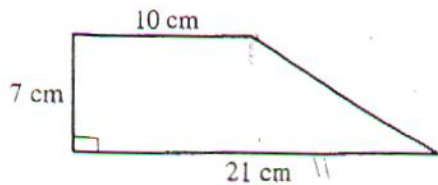
19. In the figure below, $\overline{AE}$ and $\overline{BD}$ intersect at C , and $\overline{BD}$ is perpendicular to both $\overline{AB}$ and $\overline{ED}$. The length of $\overline{AB}$ is 10 feet, the length of $\overline{BC}$ is 40 feet, and the length of $\overline{CD}$ is 120 feet. What is the length, in feet, of $\overline{DE}$?



- A. 10
B. 30
C. 35
D. 40
E. 60
20. The figure below consists of a geometric shape and a line through point P . If this figure is rotated 180° clockwise about point P , which of the following is the figure's image?
(Note: The rotation is in the plane of the figure.)



21. To the nearest centimeter, what is the perimeter of the trapezoid shown below?

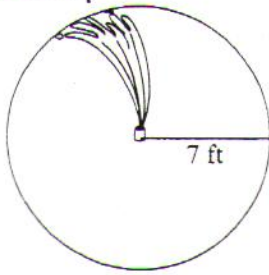


- A. 48
B. 51
C. 54
D. 56
E. 60

DO YOUR FIGURING HERE.



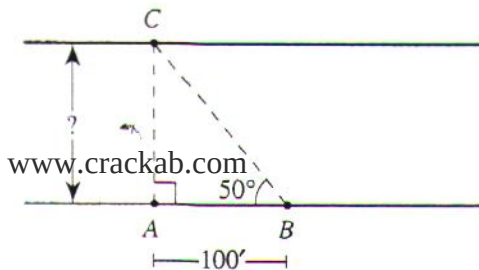
22. A rotating sprinkler sprays the entire area of a circular region with water out to 7 feet in every direction, as shown in the figure below. What is the area sprayed, to the nearest square foot?



DO YOUR FIGURING HERE.

- F. 22
G. 38
H. 44
J. 154
K. 616

23. A surveyor needs to know the width of a flood control channel. From point A , she walks 100 feet to point B and sights to point C , as shown below.



From this sighting, the surveyor finds that the measure of $\angle ABC$ is 50° . Which of the following is closest to the width, in feet, of the flood control channel?

(Note: $\sin 50^\circ \approx 0.766$
 $\cos 50^\circ \approx 0.643$
 $\tan 50^\circ \approx 1.192$)

- A. 64
B. 77
C. 119
D. 131
E. 156

24. What is the greatest common factor of the following 3 monomials?

$$20x^5y^3, x^2y^{10}, 14x^8y^8$$

- F. x^2y^3
G. $2x^8y^{10}$
H. $14x^8y^8$
J. $20x^5y^3$
K. $20x^8y^{10}$



Use the following information to answer questions 25–27.

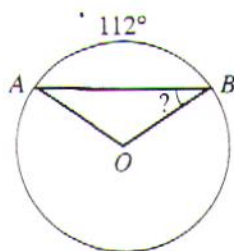
DO YOUR FIGURING HERE.

Two friends, Chris and Pat, joined a new 14-week program to lose weight through dieting and exercise. Chris set a goal to lose 10 pounds every 4 weeks, for a total of 35 pounds, while Pat's goal was to lose 7 pounds every 4 weeks, for a total of 24.5 pounds. Chris began the program weighing 200 pounds, and Pat began the program weighing 187.5 pounds.

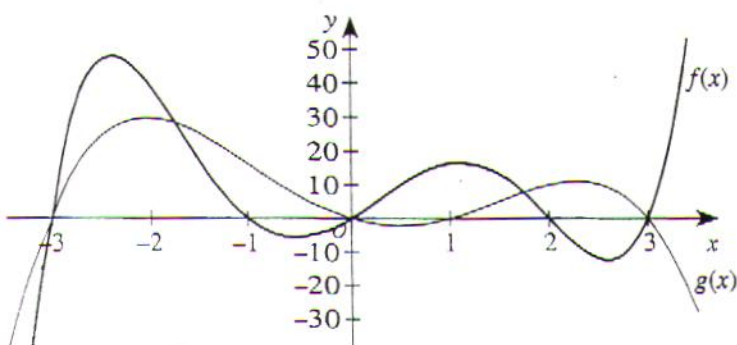
25. If both Chris and Pat met their goals exactly, how many more pounds would Chris weigh than Pat at the end of the program?
- A. 0
B. 0.75
C. 2
D. 3
E. 10.5
26. Chris lost 5 pounds during the first 2 weeks of the program, but lost only 3 pounds during the next 2 weeks. How many pounds of weight loss per week will Chris need to average during the remaining 10 weeks to meet the goal of losing 35 pounds during the program?
- F. 2.7
G. 3
H. 3.5
J. 4
K. 5
27. Suppose that Pat loses 1.75 pounds each week. At the end of week k , where $k = 1, 2, \dots, 14$, he calculates 1.75 pounds as a percent of his body weight and records the percent as $P(k)$. Which of the following best describes the values of $P(k)$?
- A. They increase as k increases from 1 to 14.
B. They decrease as k increases from 1 to 14.
C. They increase as k increases from 1 to 7 and decrease as k increases from 7 to 14.
D. They decrease as k increases from 1 to 7 and increase as k increases from 7 to 14.
E. They are constant as k increases from 1 to 14.



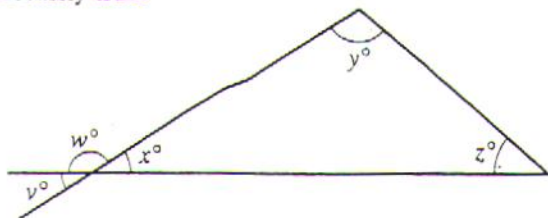
28. The circle below has its center at O , and points A and B lie on the circle. The length of $\overline{AO}$ is 4 inches, and arc $\widehat{AB}$ measures 112° . What is the measure of $\angle ABO$?



- E. 17°
 G. 34°
 H. 45°
 J. 56°
 K. 68°
29. The functions $f(x) = x(x+3)(x+1)(x-3)(x-2)$ and $g(x) = -x(x+3)(x-3)(x-1)$ are graphed below in the standard (x,y) coordinate plane. How many solutions are there to $f(x) = g(x)$?



- A. 2
 B. 3
 C. 5
 D. 6
 E. Infinitely many
30. Three lines intersect, as shown below, with angle measures as marked. Which of the following is NOT necessarily true?



- F. $v + y + z = 180$
 G. $v + w = 180$
 H. $w + x = 180$
 J. $x + z = y$
 K. $y + z = w$

DO YOUR FIGURING HERE.



31. The sum of the solutions for the equation $(x + 2)(x - 3) = 0$ is:

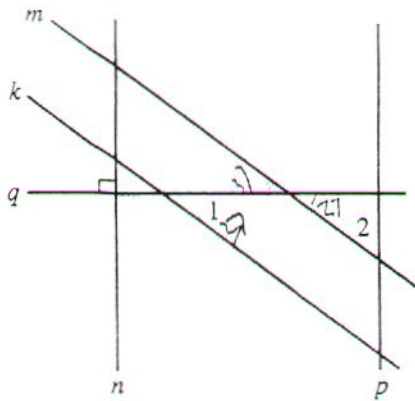
A. -6
B. -5
C. 5
D. -1
E. 1

DO YOUR FIGURING HERE.

32. If $2x - 3(2 - x) = 19$, what is the value of $2 - 3x$?

F. 73
G. 41
H. 23
J. -13
K. -73

33. In the figure below, lines k and m are parallel, lines n and p are parallel, and lines n and q are perpendicular. If the measure of $\angle 1$ is 37° , what is the measure of $\angle 2$?



A. 37°
B. 43°
C. 53°
D. 63°
E. Cannot be determined from the given information

34. What is the measure, in degrees, of an angle of $\frac{\pi}{6}$ radians?

F. 10°
G. 20°
H. 30°
J. 60°
K. 120°

35. If $2^3 \cdot 4^2 = 2^x$, then $x = ?$

A. $\frac{3}{2}$
B. 5
C. 6
D. 7
E. 12



36. What is the slope of the line $6x - 4y = 8$ in the standard (x,y) coordinate plane?

DO YOUR FIGURING HERE.

F. $-\frac{3}{2}$

G. $-\frac{2}{3}$

H. $\frac{2}{3}$

J. $\frac{3}{4}$

K. $\frac{3}{2}$

37. What is the set of all integers in $\{5, 6, 7, 8, 9\}$ that are also in the solution set of $2x + 7 < 3x + 1$?

A. $\{5, 6, 7\}$

B. $\{6, 7, 8, 9\}$

C. $\{7\}$

D. $\{7, 8, 9\}$

E. $\{9\}$

38. A rectangular painting is 22 cm wide and 28 cm tall. It will be framed with a mat. The mat forms a uniform border 4 cm wide around the painting. A piece of glass, cut to size, will cover the painting and mat exactly. What is the area, in square centimeters, of the piece of glass?

F. 148

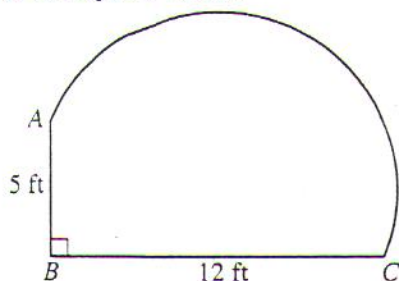
G. 464

H. 616

J. 1,080

K. 2,464

39. In the figure below, $\widehat{AC}$ is a semicircle. What is the perimeter of the figure, in feet?



A. $\frac{17}{2} + \frac{13\pi}{2}$

B. $17 + \frac{13\pi}{2}$

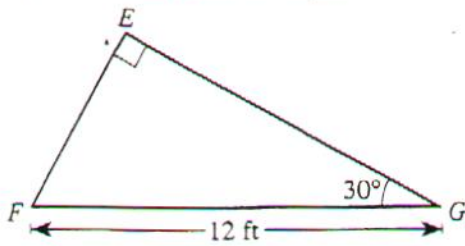
C. $17 + 13\pi$

D. $30 + 13\pi$

E. $34 + 13\pi$

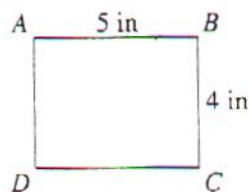


40. In the figure below, $\triangle EFG$ has angles and a side length as given. What is the measure, in feet, of $\overline{EG}$?



DO YOUR FIGURING HERE.

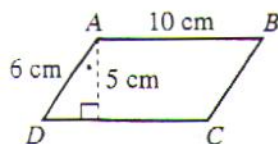
- F. $6\sqrt{3}$
 G. $6\sqrt{2}$
 H. 9
 J. 6
 K. Cannot be determined from the given information
41. A state's department of motor vehicles wants to issue licenses that contain 3 of 24 letters (the 26 letters of the alphabet except for O and I), followed by 2 of 10 digits. No single license should use any letter or digit more than once. Two such examples for licenses are AXL34 and ALX34. When the restrictions are followed, how many different licenses are possible?
- A. $26^3 \cdot 10^2$
 B. $24^3 \cdot 10^2$
 C. $22^3 \cdot 9^2$
 D. $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9$
 E. $24 \cdot 23 \cdot 22 \cdot 10 \cdot 9$
42. In rectangle $ABCD$ below, $\overline{AB}$ is 5 inches long and $\overline{BC}$ is 4 inches long. If point P lies on $\overline{AB}$, what is the area of $\triangle PCD$, in square inches?



- F. 5
 G. 9
 H. 10
 J. 20
 K. Cannot be determined from the given information
43. The 13th, 14th, and 15th terms of an arithmetic sequence are 61, 65, and 69, respectively. What are the first 2 terms of the sequence?
- A. 4, 8
 B. 9, 11
 C. 9, 13
 D. 13, 15
 E. 13, 17



44. What is the area, in square centimeters, of parallelogram $ABCD$ shown below?



- F. 30
G. 40
H. 50
J. 55
K. 60

DO YOUR FIGURING HERE.

45. What is the smallest positive number that is a multiple of both 12 and 90?

- A. 2
B. 6
C. 180
D. 360
E. 900

46. When the original price of a sweatshirt is discounted by 25% and the discounted price is then reduced by 10%, the resulting price is the original price discounted by what percent?

- F. 15%
G. 17.5%
H. 25%
J. 32.5%
K. 35%

47. In the standard (x,y) coordinate plane, approximately how many coordinate units is $(5,3)$ from $(9,-3)$?

- A. 3
B. 4
C. 7
D. 10
E. 14

48. In the standard (x,y) coordinate plane, what is the equation of the line that is parallel to the y -axis and contains the point $(-3,2)$?

- F. $x = -3$
G. $x = 2$
H. $x = 3$
J. $y = -3$
K. $y = 2$



Use the following information to answer questions 49–51.

DO YOUR FIGURING HERE.

The DigiPhone Company advertises the following calling plans:

DigiPhone Calling Plans

600 minutes* for \$ 89.99** per month
 1,000 minutes* for \$119.99** per month
 1,400 minutes* for \$149.99** per month

*The charge for each additional minute is \$0.20**.
 **Taxes are NOT included.

49. Tanisha is on the \$89.99 plan, and Suki is on the \$119.99 plan. To the nearest cent, how much more is the before-tax charge per minute for 600 minutes on Tanisha's plan than for 1,000 minutes on Suki's plan?
- A. \$0.03
 B. \$0.12
 C. \$0.13
 D. \$0.15
 E. \$0.75
50. Luis chose the \$149.99 plan. There are certain months when he uses more than 1,400 minutes of calling time. For a month in which he uses x minutes of calling time, where x is an integer greater than 1,400, which of the following expressions gives the amount, in dollars, of his bill before any taxes are added?
- F. $0.20x$
 G. $149.99 + 0.20x$
 H. $149.99 + 0.20(1,400 - x)$
 J. $149.99 + 0.20(1,400 + x)$
 K. $149.99 + 0.20(x - 1,400)$
51. In the standard (x,y) coordinate plane, the points $(600, 89.99)$, $(1,000, 119.99)$, and $(1,400, 149.99)$ are collinear. DigiPhone wants to offer a new plan that includes 400 minutes of calling time for a charge in line with those of its other plans. If $\$z$ represents the before-tax charge for this new plan, what should be the value of z so that $(400,z)$ is on the same line as the points representing the other plans?
- A. 26.67
 B. 44.99
 C. 53.33
 D. 59.99
 E. 74.99



52. Whenever $\cos x \neq 0$ and $\sin x \neq 0$, the expression $\sec^2 x + \tan^2 x$ is equivalent to:

F. $1 + \sin 2x$

G. $1 + \cos 2x$

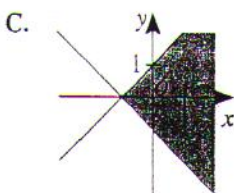
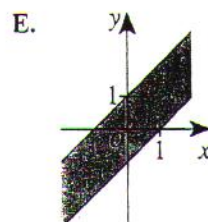
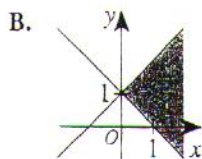
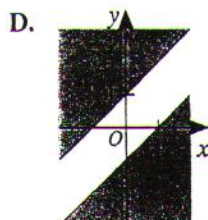
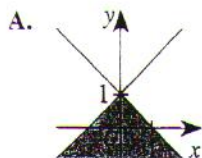
H. $\frac{1 + \cos^2 x}{\cos^2 x}$

J. $\frac{2 + \cos^2 x}{\cos^2 x}$

K. $\frac{1 + \sin^2 x}{\cos^2 x}$

DO YOUR FIGURING HERE.

53. The shaded portion of one of the following graphs is the set of all x and y such that $-1 \leq x - y \leq 1$. Which one?



54. In the standard (x, y) coordinate plane, the graph of the equation $x^2 + 3y^2 - 4x + 12y = 0$ is:

- F. a point.
G. a line.
H. a hyperbola.
J. a circle.
K. an ellipse.



55. When x is divided by 8, the remainder is 5; and when y is divided by 6, the remainder is 1. If each of x and y is at least 20, what is the minimum value of $x + y$?

A. 40
B. 46
C. 52
D. 54
E. 60

DO YOUR FIGURING HERE.

56. The numbers a and b are real numbers and $|a| \cdot b = a \cdot |b|$. Which of the following *must* be true about the relationship between a and b ?

F. $a = -b$
G. $a = b$
H. $ab \leq 0$
J. $ab \geq 0$
K. $a = 0$ or $b = 0$

57. If k is an integer and $k > 2$, which of the following expressions is equivalent to $\frac{(k+1)!(k-1)!}{(k!)^2}$?

(Note: For any positive integer n , $n! = (n)(n-1)(n-2) \cdots (2)(1)$. Hence $5! = (5)(4)(3)(2)(1) = 120$.)

A. $\frac{k+1}{k}$
B. $k^2 - 1$
C. -1
D. 0
E. 1

58. The lengths, in inches, of all 3 sides of a triangle are positive integers, and 1 side is 7 inches long. If the area of this triangle is positive, what is the smallest possible perimeter for this triangle, in inches?

F. 9
G. 10
H. 14
J. 15
K. 21

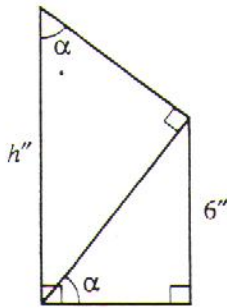
59. The square of a certain positive integer is k . Which of the following expressions is equivalent to the square of the next integer greater than that certain positive integer?

A. $\sqrt{k+1}$
B. $k+1$
C. k^2+1
D. $(k+1)^2$
E. $(\sqrt{k}+1)^2$



60. In the figure below, $\sin \alpha = \frac{4}{5}$. What is the value of h ?

DO YOUR FIGURING HERE.



F. $\frac{24}{5}$

G. $\frac{34}{5}$

H. $\frac{36}{5}$

J. $\frac{75}{8}$

K. $\frac{15}{2}$

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the short story "The Behavior of the Hawkweeds" by Andrea Barrett (©1996 by Andrea Barrett). Gregor Mendel was a nineteenth-century botanist, teacher, and monk who discovered the mathematical basis for the study of genetics.

For thirty years, until he retired, my husband stood each fall in front of his sophomore genetics class and passed out copies of Gregor Mendel's famous paper on the hybridization of edible peas. This paper was a
5 model of clarity, Richard told his students. It represented everything that science should be.

Richard paced in front of the chalkboard, speaking easily and without notes. Like the minor evolutionist Robert Chambers, he had been born hexadactylic; he
10 was sensitive about his left hand, which was somewhat scarred from the childhood operation that had removed his extra finger. And so, although Richard gestured freely he used only his right hand and kept his left in his pocket. From the back of the room, where I sat
15 when I came each fall to hear him lecture, I could watch the students listen to him.

After he passed out the paper, Richard told the students his first, conventional version of Gregor Mendel's life. Mendel, he said, grew up in a tiny village in the
20 northwestern corner of Moravia, which was then a part of the Hapsburg Empire and later became part of Czechoslovakia. When he was twenty-one, poor and desperate for further education, he entered the Augustinian monastery in the capital city of Brunn, which is
25 now called Brno. He studied science and later taught at a local high school. In 1856, at the age of thirty-four, he began his experiments in the hybridization of the edible pea. For his laboratory he used a little strip of garden adjoining the monastery wall.

Over the next eight years Mendel performed hundreds of experiments on thousands of plants, tracing the ways in which characteristics were passed through generations. Tall and short plants, with white or violet
30 flowers; peas that were wrinkled or smooth; pods that were arched or constricted around the seeds. He kept meticulous records of his hybridizations in order to write the paper the students now held in their hands. On a clear, cold evening in 1865, he read the first part of this paper to his fellow members of the Brunn Society

40 for the Study of Natural Science. About forty men were present, a few professional scientists and many serious amateurs. Mendel read to them for an hour, describing his experiments and demonstrating the invariable ratios with which traits appeared in his hybrids. A month
45 later, at the Society's next meeting, he presented the theory he'd formulated to account for his results.

Right there, my husband said, right in that small, crowded room, the science of genetics was born. Mendel knew nothing of genes or chromosomes or
50 DNA, but he'd discovered the principles that made the search for those things possible.

"Was there applause?" Richard always asked at this point. "Was there a great outcry of approval or even a mutter of disagreement?" A rhetorical question;
55 the students knew better than to answer.

"There was not. The minutes of that meeting show that no questions were asked and no discussion took place. Not one person in that room understood the significance of what Mendel had presented. A year later,
60 when the paper was published, no one noticed it."

The students looked down at their papers and Richard finished his story quickly, describing how Mendel went back to his monastery and busied himself with other things. For a while he continued to teach and
65 to do other experiments; he raised grapes and fruit trees and all kinds of flowers, and he kept bees. Eventually he was elected abbot of his monastery, and from that time until his death he was occupied with his administrative duties. Only in 1900 was his lost paper rediscovered and his work appreciated by a new generation of
70 scientists.

When Richard reached this point, he would look toward the back of the room and catch my eye and smile. He knew that I knew what was in store for the
75 students at the end of the semester. After they'd read the paper and survived the labs where fruit flies bred in tubes and displayed the principles of Mendelian inheritance, Richard would tell them the other Mendel story. The one I told him, in which Mendel is led astray by a
80 condescending fellow scientist and the behavior of the hawkweeds. The one in which science is not just unappreciated, but bent by loneliness and longing.

1. Which of the following statements best describes the qualities of Mendel's scientific work as it is depicted in the passage?
 - A. Using a laboratory supplied by the monastery where he lived, he carefully conducted experiments designed by professional scientists and serious amateurs.
 - B. His research on edible-pea plants was systematic and thorough, and he recorded and presented his findings with great precision and care.
 - C. He carefully investigated numerous characteristics that could be inherited by edible-pea plants in a search for an explanation of genes, chromosomes, or DNA.
 - D. His research interests led him to use thousands of edible-pea plants as the basis for his many experiments, some of which he kept records of in order to write his paper.
2. Which of the following questions about Mendel's life is NOT answered in the passage?
 - F. How did he become interested in the study of science?
 - G. Why did he enter the Augustinian monastery?
 - H. Where did he conduct his hybridization experiments?
 - J. What were some of the pea characteristics he studied?
3. Richard most nearly describes Mendel as being:
 - A. persistent, egotistical, and educated.
 - B. argumentative, controlling, and hexadactylic.
 - C. ambitious, kindly, and wealthy.
 - D. curious, methodical, and unappreciated in his lifetime.
4. Which of the following statements best summarizes Richard's view of the importance of Mendel's work?
 - F. Mendel's published paper was more significant than were his many experiments with edible peas.
 - G. Mendel did not know about genes, chromosomes, or DNA, which limits his work's usefulness to modern scientists.
 - H. Mendel's revolutionary discoveries, published in the paper Richard described, pale in significance to his later work.
 - J. Mendel, through his solitary studies, paved the way for major advances in science.
5. As it is presented in the passage, the relationship between Richard and the narrator appears to be based on:
 - A. collegiality and sarcasm.
 - B. shared humor and obedience.
 - C. affection and shared interests.
 - D. watchfulness and one-upmanship.
6. As they are presented in the fourth paragraph (lines 30–46), Mendel's findings can best be described as:
 - F. promising but inconclusive, since the ratios he discovered held true for only some characteristics in edible-pea plants.
 - G. useful and revealing, since he had found rules by which the transmission of traits occurs in hybrids of edible-pea plants.
 - H. suggestive but tentative, since he was unable to develop a reasonable theory to account for his data.
 - J. clear and important, since his research almost immediately led to other important scientific discoveries.
7. The way in which Richard asks questions in the sixth paragraph (lines 52–55) and answers them in the seventh paragraph (lines 56–60) is most likely meant to emphasize how:
 - A. obvious it is that the members of the Brünn Society secretly disagreed with Mendel's conclusions.
 - B. embarrassing it is that Mendel, a future abbot, spent his time writing an incomprehensible paper.
 - C. surprising it is that Mendel's groundbreaking paper received so little attention when it was first presented.
 - D. humorous it is that no one approved of or disagreed with Mendel's conclusions.
8. It can reasonably be inferred that Richard considers the details about Mendel's life in the eighth paragraph (lines 61–71) to be:
 - F. worth mentioning, because Mendel's work as an abbot is evidence of the respect others felt for him.
 - G. interesting in what they reveal about Mendel's later experiments with grapes, fruit trees, and flowers.
 - H. of less interest, because much of Mendel's life at that time involved administrative work rather than science.
 - J. disturbing, because such sketchy information is available about the rest of Mendel's life.
9. It can most reasonably be inferred from the passage that the narrator uses the word *survived* (line 76) with regard to the students' lab work to suggest which of the following?
 - A. The narrator knows that the students will find the lab work difficult, time-consuming, or both.
 - B. Most students drop the course before starting the lab work because they are afraid of the experiments.
 - C. Richard dislikes assigning lab work and therefore makes it as difficult to perform as possible.
 - D. The narrator realizes the lab work is dangerous but feels it can be accomplished under the guidance of a good teacher.

10. Based on the passage, "the other Mendel story" (line 78) is one that the narrator:

F. has yet to tell to Richard.
 G. describes only in general terms.
 H. enjoys telling to Richard's students.
 J. claims is not historically accurate.

Passage II

SOCIAL SCIENCE: This passage is adapted from the biography *Marian Anderson: A Singer's Journey* by Allan Keiler (©2000 by Allan Keiler).

On December 30, 1935, manager Sol Hurok presented singer Marian Anderson in New York's Town Hall. She had been away from the United States for nearly two and a half years and Hurok, uncertain of the audience she would draw, chose the smaller theater for her homecoming rather than the more prestigious Carnegie Hall. One can only wonder whether he foresaw the extraordinary potential of the contralto singer whom he had agreed to manage, or the remarkable relationship they would enjoy.

By 1939, after only five concert seasons, Anderson had more than duplicated in America what she had achieved in Europe. In America, her fees had risen from \$400 to \$2,000 for most performances. With more engagements for Anderson than she could comfortably fulfill, Hurok's job was now "not to seek dates for Marian, but to winnow out the most attractive ones each season." To meet the repertoire needs of constantly expanding seasons, which included yearly appearances in many cities, she had to change programs more rapidly. In developing her repertoire, she had the help of a network of teachers on both sides of the Atlantic. She added an abundance of new material, exploring the more unusual, even esoteric byways of the vocal literature. Her programs had a striking cosmopolitanism, which she liked to emphasize particularly in her Carnegie Hall recitals. For her first recital there in 1939, she chose music by Spanish composers, a group of French chansons, a Tchaikovsky aria, music by American composers, and spirituals.

In only five seasons with Hurok, Anderson had become an indelible part of the American musical scene. Occasionally a critic preferred to maintain some appearance of sobriety in response to an Anderson concert. Olin Downes, for example, never comfortable with extravagant praise, held to his gravity of manner in reviewing Anderson's first New York recital in 1939: "Miss Anderson sings music by classic masters, not as a lesson learned, or a duty carefully performed, but as an interpreter who has fully grasped and deeply felt the import of the song." For the rest, critics threw caution to the winds, outdoing each other in their efforts to capture the excitement generated by Anderson's appearances. From New York to Buffalo to Milwaukee, Anderson was rewarded with a triumphal chorus of brilliant reviews.

Of course, there was much more to Anderson's success than voice and art. In her every appearance, the unique qualities of her character and personality made themselves felt just as strongly as her technique and interpretations. Perhaps, in the end, what touched audiences most deeply was the provocative contradiction between the effortless warmth, sincerity, and dignity Anderson radiated and the quality of aloofness that was always present. The critic Glenn Dillard Gunn, after one of her Washington recitals, provided a moving portrait of Anderson's compelling hold on the imagination of her audiences: "Miss Anderson's power to move her listeners as can no other singer of her generation is not made less by the simple, almost unsmiling dignity which clothes her like a garment. She holds herself aloof and doubtless with intention. Her eyes closed, her face expressionless, she retires into her world of song, inviting us to enter it, but remaining remote, not sharing her personality. Even in her moment of triumph the barrier is there."

Radio helped introduce Anderson's voice across America. With the waning of the economic hardships left behind by the Depression, radio opportunities for classical artists increased significantly. During the late thirties, her voice could be heard on such programs as the General Motors Hour, the Magic Key program and the Ford Hour. Her striking voice, never more opulent than in those years; the range of her programs, which included, in addition to spirituals, more popular numbers than she normally sang on concert programs; and the charm and radiance of her personality, so intimately bound up with her singing, made her an extraordinarily popular radio performer.

In the black community, Anderson's accomplishments as a singer and the sense of pride and purpose she inspired brought her many honors. In January 1939, the NAACP announced that it would award Anderson the Spingarn Medal, given annually by its president. Joel Spingarn, "for the highest or noblest achievement by an American Negro during the preceding year or years." Since the inception of the Spingarn award in 1915, only two other musicians, Roland Hayes and Harry T. Burleigh, had been so honored.

11. The main purpose of the passage is to:

A. reconcile several differing interpretations of Anderson's return to the United States.
 B. offer a point-by-point comparison between Anderson's European and American careers.
 C. portray Anderson's relationship with Hurok and his role in expanding and improving her repertoire.
 D. describe one successful period in Anderson's career and the nature of her personality and music.

12. Based on the passage, one difference between an Anderson concert at Carnegie Hall and an Anderson concert on the Magic Key radio program would most likely be that:
- F. one would hear more popular songs on the Magic Key program.
 - G. one would hear spirituals on the Magic Key program.
 - H. the Carnegie Hall concert would be limited to songs by European composers.
 - J. the Carnegie Hall concert would be less cosmopolitan in content.
13. All of the following statements about Anderson's career by the end of 1939 are true according to the passage EXCEPT that she had:
- A. been told she would receive the Spingarn Medal.
 - B. become a highly popular radio performer.
 - C. begun to plan a new tour of Europe.
 - D. given her first Carnegie Hall recital.
14. The first paragraph makes it clear that as a manager, Hurok:
- F. felt the Town Hall concert was an overly conservative choice by Anderson.
 - G. knew Anderson deserved the biggest possible audience for her homecoming.
 - H. could see how famous Anderson would eventually become.
 - J. planned Anderson's reintroduction to American audiences cautiously.
15. As described in the fourth paragraph (lines 47–66), a contradiction in Anderson is that she could:
- A. sing beautiful songs with great warmth yet be cold and insincere as a person.
 - B. interest audiences in her technique and interpretations but not her personality.
 - C. powerfully move her listeners in concerts but not in recordings.
 - D. seem warm and sincere yet at the same time distant during a performance.
16. If the fifth paragraph (lines 67–79) were omitted, readers would not know that:
- F. Anderson's popularity was increased by radio.
 - G. Anderson performed wide-ranging programs.
 - H. Anderson was popular during the late thirties.
 - J. Anderson's voice was more opulent when she sang on radio.
17. As it is used in line 17, the phrase *winnow out* most nearly means:
- A. eliminate.
 - B. select only.
 - C. solicit.
 - D. bargain for.
18. According to the author of the passage, which of the following people tried to show restraint in reviewing an Anderson concert?
- F. Hurok
 - G. Downes
 - H. Gunn
 - J. Spingarn
19. The main point of the quotation in lines 38–41 is that Anderson:
- A. was diligent in performing music by leading composers.
 - B. truly understood and was able to personalize well-regarded songs.
 - C. had learned her lessons early when it came to grasping classical music.
 - D. deeply appreciated classical music and enjoyed hearing it performed.
20. The quotation in lines 58–66 characterizes Anderson's aloofness as:
- F. deliberate.
 - G. accidental.
 - H. regrettable.
 - J. misunderstood.

Passage III

HUMANITIES: This passage is adapted from the preface to *On Native Grounds: An Interpretation of Modern American Prose Literature* by Alfred Kazin (©1995 by Alfred Kazin).

I began *On Native Grounds* on a kitchen table in Brooklyn, 1938, and completed it in Long Island City, 1942, expecting a call from my draft board at any minute. The dates are essential to any understanding of the book, to its survival for over fifty years and its continued influence. There is no excitement for a writer like that of living in rebellious times. At least before World War II broke out, my work in progress was very much the product of and a response to the social crises of the '30s. The massive breakdown of the American economy in the depression was the greatest national crisis after the Civil War, and I lived in its very midst, tossed up and down in the stormy ocean of the times by the suffering of my unemployed working-class parents, the mass social protests all over the country, the triumph of Fascism in Germany, Italy, and Spain, and the extremism in America itself of Communist and Fascist ideologies in violent conflict.

What Walt Whitman wrote about the Civil War in "Drum-Taps" could have been said of the '30s:

Long, too long America,
Traveling roads all even and peaceful you learn'd from joys and prosperity only,
But now, ah now, to learn from crises of anguish, advancing, grappling with direst fate and recoiling not,
And now to conceive and show to the world what your children enmasse really are . . .

A history of modern American prose literature begun in such a period and continued out of a sense of social crisis during the great global war against Fascism! The literary significance of this is that I believed in what essayist William Hazlitt called the "spirit of the age"—meaning that *this* age we were living in had a character all its own and could be related to other ages and periods, thus constituting a historical scene in which a period was known through its writers and its writers through their period.

Of all my books, *On Native Grounds* was the easiest to write. I felt what I have never felt since 1945—that the age was wholly with me, that I was appealing to "the spirit of the age," that the writers as characters in my book were friends and the most encouraging people in the world to write about. I was writing literary *history*, a genre long abandoned by critics and now suspect (history can no longer be characterized and summed up as confidently as it was in the '30s and early '40s by the young man who wrote this book). This means that I saw connections everywhere between history and literature. I saw connections between the writers themselves as fellow-spirits and artists relating to the pressures of American life.

My subject was the emergence of the "modern" in an American literature obviously unsettled by relentless

new forces in every sphere: social, intellectual, and religious. My perspective, so natural in the turbulent '30s, was based on a spirit of social protest I shared with almost every writer in my book.

There was nothing strange or unexpected in 1938 about my being both critical of "the system" and crazy about the country. What drew me to the serious study of American literature within a historical context was the narrative it suggested on every hand. America from its beginnings as "our rising empire" (George Washington) embodied a purposeful form of historical movement, unprecedented on such a continental scale, that cried out to be written as a great story. In the background of the particular story I was writing was the sense, which was everywhere at the end of the nineteenth century, of a new age. What struck me from the first was the astonishment with which American writers confronted situations as new to themselves as to the Europeans who were often reading about America for the first time.

What gave me the confidence at twenty-three to begin a book like this? The age, the insurgency of the times, but above all *On Native Grounds* represents my personal discovery of America. The first native son in my immigrant family, brought up in a Brooklyn ghetto by parents whose harshly enclosed lives never gave them a chance even to learn English, I was crazy about the America I knew only through books. And it was such an idealistic America, defined by its purest spirits, from Audubon and Jefferson to Emerson and Thoreau, to the Lincoln who had saved the Union, to the great democrats of philosophy John Dewey and William James, and to the Willa Cather, Theodore Dreiser, Sherwood Anderson, and Carl Sandburg who brought home the Middle West to me as the valley of democracy and the fountainhead of hope.

21. To which facet of American life does the passage's author make the fewest direct references in the passage?
 - A. Military
 - B. Social
 - C. Religious
 - D. Literary
22. According to the passage, when he started *On Native Grounds*, the passage's author was:
 - F. twenty-three and the year was 1938.
 - G. in his thirties and the depression had just ended.
 - H. eighteen and the year was 1938.
 - J. twenty and America had just entered World War II.

23. All of the following are details the passage's author mentions about his parents EXCEPT that they:
- A. did not speak English.
 - B. were unemployed during the depression.
 - C. were proud of their son's accomplishments.
 - D. raised a family in Brooklyn, New York.
24. The poem quoted in lines 21–27 is best described as:
- F. Kazin's poem written as a portrait of America in transition in the 1930s.
 - G. Kazin's poem bemoaning the direction taken by America after the Civil War.
 - H. Whitman's poem, which Kazin uses as an example of patriotic but unpopular verse.
 - J. Whitman's poem, which Kazin presents as embodying the spirit of two different eras.
25. It can reasonably be inferred from the fourth paragraph (lines 38–51) that when he wrote *On Native Grounds*, the passage's author felt his position with regard to the times was that of:
- A. an outsider wanting a way to get involved.
 - B. a person completely in tune with the times.
 - C. a youth impatient with the times' outdated traditions.
 - D. an activist capable of perfecting society.
26. It can reasonably be inferred that when the passage's author says his perspective was *natural* (line 55), he means it was:
- F. free of the influences of academic training.
 - G. based in a respect for the environment.
 - H. a product of times characterized by insurgency.
 - J. an outcome of being raised to value literature.
27. When the passage's author mentions "the valley of democracy" (line 87), he is referring to the:
- A. Union, as preserved by Lincoln at the time of the Civil War.
 - B. New World, as described by Whitman in poetry the author read as a child.
 - C. Middle West, as portrayed by Cather, Dreiser, Anderson, and Sandburg.
 - D. United States, as perceived by the author's idealistic parents before they immigrated to New York.
28. The passage's author states that compared to writing other books, writing *On Native Grounds* was:
- F. more challenging.
 - G. more disillusioning.
 - H. less time-consuming.
 - J. less difficult.
29. The passage's author states that what drew him to the serious study of American literature within a historical context was the:
- A. narrative it suggested on every hand.
 - B. breakdown of the U.S. economy.
 - C. political power of Whitman's poetry.
 - D. powerful novels written by historians.
30. One of the discoveries the passage's author says he made when writing *On Native Grounds* was that the American writers he was examining reacted to the American experience with:
- F. profound gratitude.
 - G. genuine astonishment.
 - H. cautious approval.
 - J. utter disbelief.

Passage IV

NATURAL SCIENCE: This passage is adapted from *The Earth Dwellers: Adventures in the Land of Ants* by Erich Hoyt (©1996 by Erich Hoyt).

In the steamy first light of the tropical rain forest in Costa Rica, life is buzzing. The forest echoes with the whooping calls of the oropendolas [tropical birds]. At lower volume but ever more persistent is the hissing of thousands of cicadas and other insects. In the distance howler monkeys are on dawn patrol.

No matter the racket, nocturnal animals are winding up their day and preparing to sleep, while those that need light to hunt are just awake and looking for breakfast. Still others—including some of the insects—have been working all night and will continue into the day.

From overhead a shaft of light beams down on a scene of frenetic activity. A worker ant—a foraging scout, female, as are all ant workers—stands on the leaf of a low-growing bush. Her hind legs are digging in, her head is down, hard at work. Her serrated mandibles, her jaws, are moving, sawing through a leaf. The air is pungent with leaf sap. As it drips from the leaf, she stops to lick a drop or two for refreshment. Two seconds later she's back at work.

As she saws, the sharp blade of her mandibles cuts through the leaf tissues, and her round head bobs up and down with the rhythm of a woodcutter. Slowly she begins to pivot on her hind legs to inscribe a curved line. She turns steadily, finally coming all around in a nearly perfect circle. She nudges the leaf panel to break the final perforations, and it falls to the ground. She jumps down, squats over the leaf piece, and marks it with a drop of liquid from the underside of her rear abdomen, her gaster. This marking will make the piece more attractive to the others in the colony. It also contributes enzymes that will eventually help break down the leaf. Then she grabs the leaf piece with her mandibles, hoisting it high so that it almost rests on her antennae like a floppy, oversized hat. She walks a few feet into the full sunlight of the forest clearing and stops. Her leafy hat is heavy.

In the speckled light of the clearing, the scout's entire body, seen in profile, glows reddish brown—except for the protruding black eyes. She has the distinctive spines of a leafcutter, or parasol, ant. They look like thorns rising from her back. She is covered in fine, tiny hairs from her head to her rear feet. The two antennae droop under the weight of her green load—the piece of leaf is three times her body mass. She struggles to press the green prize up high. And then another, much smaller leafcutter ant appears out of nowhere, climbs aboard the leaf piece, and clings to one end of it. She is less than a quarter the size of the scout, her nestmate. This smaller worker keeps alert for enemies. She, too, marks the leaf piece with a drop or two from her abdomen, contributing more enzymes to help break down the leaf.

Today the scout has what may well be a new leaf species—an untried plant for her colony. Her leaf-prize thrives on a few low-lying bushes at the edge of a clearing, some distance from the nest. It is an exotic plant—an introduced (that is, not native) species, perhaps a plant that escaped from a nearby farmer's field. As an exotic, it has fewer of the built-in defenses that many indigenous plants have evolved through living among insects for millions of years. Therefore, herbivorous insects, mammals, and other animals may be able to make easier use of it.

In a survey of forty-two plant species from the forests of Costa Rica, 75 percent contained terpenoids, steroids, waxes, and other defense compounds that repel leafcutters. Many tropical forest leaves are tough and have sharp spines that discourage monkeys and other animals from eating them. But the persistent leafcutters aren't much bothered. Their razor-sharp mandibles can slice through some of the fleshiest leaves and even twigs and stems. With regard to defense compounds, the leafcutters are selective about what they cut, but in species-rich tropical forests they have a lot of choice. Surprisingly, the rarity or commonness of the plant species has no real bearing on its appeal to the prospective leafcutter scout. Neither is proximity a crucial factor; leafcutters will walk a few hundred feet to a favored plant species. Researchers estimate that the tribe of gardening ants, which leafcutters belong to, cuts 12 to 17 percent of the leaves and flowers produced in the forests they inhabit. On one island in Panama, leafcutters annually cut almost 250 pounds of foliage per acre. To put it in terms of a colony, every day a single mature colony uses as much vegetation as a cow.

31. Which of the following characterizations is accurate about leafcutter ants as depicted in the passage?
- A. Hardworking and relatively particular about which plants they cut
 - B. Determined and most active during the early rainy season
 - C. Fragile and covered with fine, tiny thorns from head to rear
 - D. Most active during the night and able to use their mandibles rhythmically
32. It can most reasonably be inferred from the passage that the leafcutter ant's primary reason for cutting leaves is to provide:
- F. camouflage from predators such as hunting ants.
 - G. a platform for another, smaller leafcutter ant to ride on.
 - H. protection for itself from drowning in rain or burning in the sun.
 - J. nourishment for itself and the rest of the leafcutter ant colony.

33. According to the passage, the primary purposes of the leafcutter ant's gaster liquid are to:
- A. make leaf cutouts appealing to other leafcutter ants in the colony and to begin to break down the leaf chemically.
 - B. make leaf cutouts sticky so they are easier to carry and to provide nourishment to the leafcutter ants.
 - C. repel predators and other animals and to mark the leafcutter ant's trail through the forest so other ants can follow.
 - D. enable the leafcutter ant to cut through leaf tissues and to break the final perforations of the leaf panel so the panel can fall to the ground.
34. In line 7, the word *racket* most likely refers to all of the following EXCEPT the:
- F. hissing of thousands of cicadas and other insects.
 - G. sawing through leaf tissue by the leafcutter ant.
 - H. whooping calls of the oropendolas.
 - J. howler monkeys on dawn patrol in the distance.
35. Which of the following questions can be answered by information provided in the passage?
- A. Why do so many more species of animals live in the rain forest than in other parts of the world?
 - B. How do leafcutter ants use their hind legs in the leafcutting process?
 - C. By the time the leafcutting process has been completed, how much leaf sap has the ant consumed?
 - D. How long does it take for the enzymes from the leafcutter ant's gaster to break down a leaf piece?
36. It can reasonably be inferred from the passage that leafcutter ants pay the most attention to which one of the following criteria when selecting plants whose leaves they will cut?
- F. The plant must be free of sharp spines that could injure the leafcutter ant's mandibles.
 - G. The plant must be within a fifty-foot radius of the colony's nest.
 - H. The plant must be free of chemical defense compounds such as steroids.
 - J. The plant must be a member of one of the common species of tropical plants.
37. It can reasonably be inferred from the second paragraph (lines 7–11) that:
- A. sleep habits of nocturnal animals are determined by the amount of noise in the tropical rain forest.
 - B. leafcutter ants and other insects are the prey of howler monkeys looking for breakfast.
 - C. little seems to disturb the various morning behaviors of animals in the rain forest.
 - D. most tropical rain forest animals need light to hunt.
38. According to the passage, leafcutter ant scouts are like all other worker ants in a leafcutter colony in which one of the following ways?
- F. They make no noise.
 - G. They are female.
 - H. They cut any vegetation.
 - J. They are active during the night.
39. According to the passage, the leaf being sawed and then carried by the leafcutter ant has which one of the following characteristics?
- A. It contains pungent sap.
 - B. It has a serrated stem.
 - C. It contains terpenoids.
 - D. It has fragile midribs.
40. Which of the following statements about leafcutter ants is supported by details in the passage?
- F. A leafcutter ant produces only enough gaster liquid to mark one leaf piece in the leafcutting process.
 - G. A leafcutter ant scout is capable of carrying loads several times its body mass.
 - H. Leafcutter ants are less likely than other animals to accelerate the evolution of plant chemical defenses in the rain forest.
 - J. As light increases in the rain forest, the leafcutter ant scout's reddish brown glow intensifies.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.

SCIENCE TEST

35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage 1

Io, one of Jupiter's moons, was studied with 2 spacecraft instruments that can detect visible radiation or thermal infrared (TIR) radiation (see Table 1).

Table 1	
Instrument	Minimum surface temperature (K) needed for detection by instrument
NIMS	180
SSI	700

The instruments located volcanic regions called *hot spots* on *Io*. Hot spots are areas where molten material at or above 400 K is ejected. Three studies of *Io*'s hot spots were performed.

Study 1

The instruments on the spacecraft orbiting *Io* recorded the locations of all hot spots. Figure 1 shows the latitudes of those hot spots in both hemispheres.

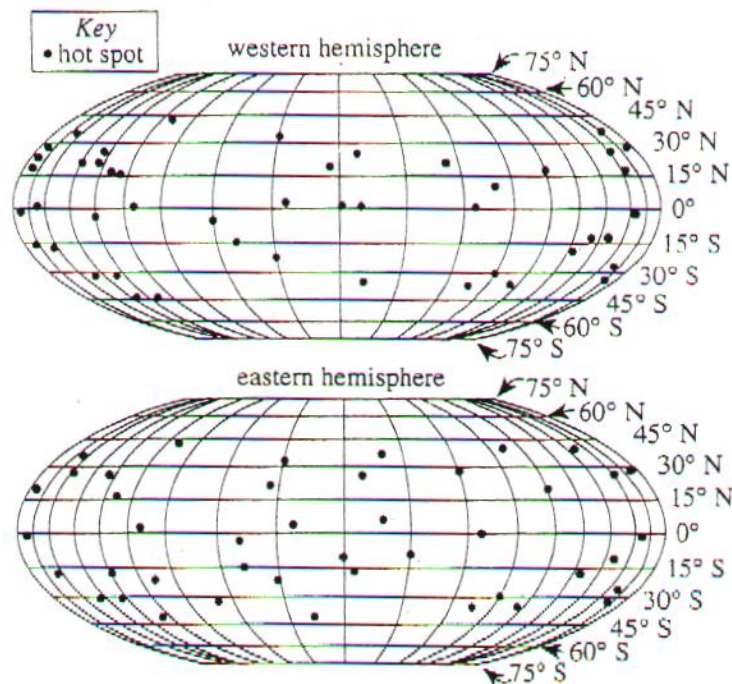


Figure 1

Study 2

The intensity of the emitted radiation from each hot spot was used to estimate the hot spot's area and average temperature. The hot spots were grouped into 2 categories: *Type I hot spots* having average temperatures below 500 K and *Type II hot spots* having average temperatures above 500 K. Type II hot spots emit higher intensity radiation than do Type I hot spots. The results for 10 hot spots are shown in Table 2.

Table 2			
	Hot Spot	Estimated average temperature (K)	Estimated area (km ²)
Type I hot spots	A	459	32
	B	485	15
	C	434	30
	D	384	192
	E	458	11
Type II hot spots	V	1,660	0.002
	W	1,525	0.02
	X	1,110	0.87
	Y	1,275	0.58
	Z	1,620	0.006

Study 3

Scientists believe that each of Io's hot spots ejects 1 of 3 types of molten material: molten sulfur, material similar in composition to basalt lava, or material similar in composition to ultramafic lava. Basalt lava and ultramafic lava have been ejected from Earth's volcanoes. Table 3 gives the melting point of sulfur and the 2 types of lava, and the boiling point of sulfur, all at a pressure of 1 atmosphere (atm).

Table 3	
	Temperature (K)
Sulfur melting point	373
Sulfur boiling point	715
Basalt lava melting point	1,000
Ultramafic lava melting point	1,400

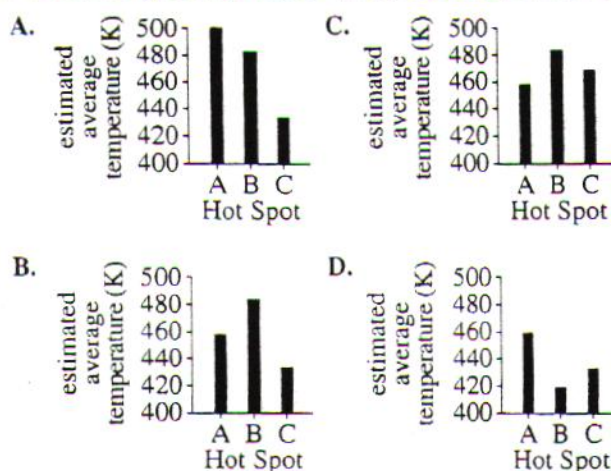
1. It was hypothesized that the vast majority of Io's hot spots would be found in Io's western hemisphere. Is this hypothesis supported by the results of Study 1?

- A. No, because similar numbers of hot spots are found in each hemisphere.
- B. No, because very few hot spots are found in the eastern hemisphere.
- C. Yes, because similar numbers of hot spots are found in each hemisphere.
- D. Yes, because very few hot spots are found in the eastern hemisphere.

2. According to the results of Study 1, hot spots on Io are located exclusively:

- F. at the equator (0° latitude).
- G. between 15° N and 15° S latitudes.
- H. between 30° N and 30° S latitudes.
- J. between 45° N and 45° S latitudes.

3. Which of the following graphs best represents the estimated average temperatures of Hot Spots A, B, and C?



4. Assume that the hot spots listed are typical of all Type I and Type II hot spots on Io. On the basis of this assumption and the results of Study 2, one would most reasonably generalize that a Type II hot spot would have an estimated area that is:

- F. smaller than the estimated area of a Type I hot spot.
- G. the same as the estimated area of a Type I hot spot.
- H. larger than the estimated area of a Type I hot spot.
- J. in some cases smaller and in some cases larger than the estimated area of a Type I hot spot.

5. Another hot spot, which had an estimated average temperature of 1,350 K, was discovered on Io. Based on the results of Study 2, the estimated area of this newly discovered hot spot would be:

- A. less than 0.02 km².
- B. between 0.02 km² and 0.58 km².
- C. between 0.58 km² and 0.87 km².
- D. greater than 0.87 km².

6. Given the information about how Io's hot spots were studied, which of the following statements about the use of the instruments is most likely true?

- F. The instruments could detect emitted radiation of any wavelength.
- G. The instruments were used primarily to detect emitted ultraviolet radiation.
- H. The instruments did not directly sample the molten materials.
- J. The instruments directly sampled the molten materials.

Passage II

When 2 substances are combined to form a mixture, various properties emerge that are different from those of the original components. Table 1 lists the mass, volume, and *density* (mass divided by volume) of mixtures of powdered sucrose and powdered sodium bromide at 25°C, and the masses of the substances that were combined to form the mixtures. Figure 1 shows how mass changes with volume for powdered sucrose and powdered sodium bromide at 25°C. Figure 2 compares the final volume of various ethanol-water mixtures to the initial volumes of the 2 pure liquids that were combined to form the mixture at a specific temperature.

Table 1

Mass of sucrose (g)	Mass of sodium bromide (g)	Mass of mixture (g)	Volume of mixture (mL)	Density of mixture (g/mL)
15.00	3.00	18.00	10.58	1.70
12.00	6.00	18.00	9.60	1.88
9.00	9.00	18.00	8.68	2.07
6.00	12.00	18.00	7.78	2.31
3.00	15.00	18.00	6.88	2.62

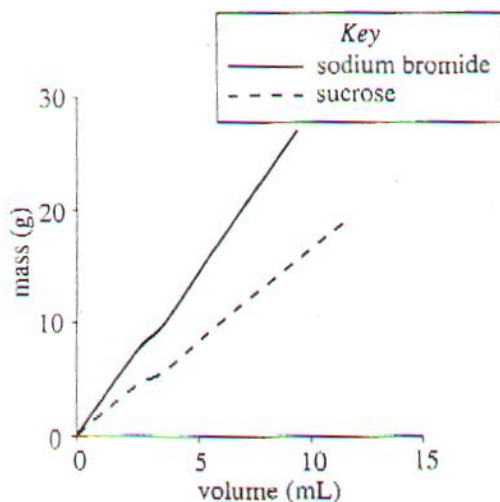


Figure 1

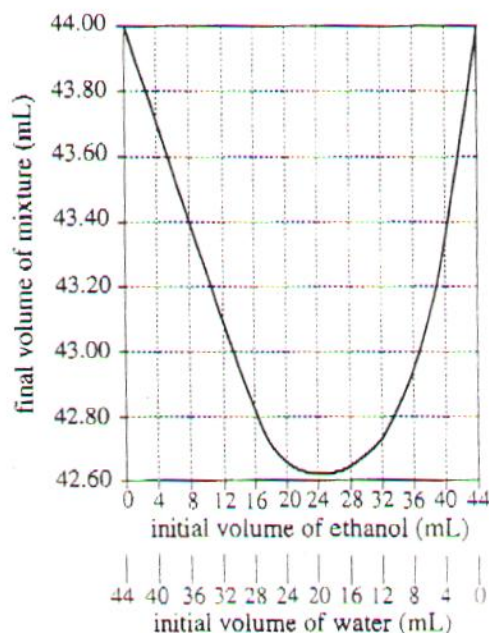


Figure 2

Table and figures adapted from the Chemical Bond Approach Project, *Chemical Systems*. ©1964 by Earlham College Press, Inc.

- According to Figure 2, as the initial volume of ethanol increased from 0 mL to 44 mL, the final volume of the mixture:
 - increased only.
 - increased, then decreased.
 - decreased, then increased.
 - remained constant.
- According to Figure 2, when 36 mL of ethanol was mixed with 8 mL of water, the final volume of the resulting mixture was closest to:
 - 42.60 mL.
 - 42.95 mL.
 - 43.38 mL.
 - 44.00 mL.
- Based on the data in Table 1, if 4 g of powdered sucrose were mixed with 14 g of powdered sodium bromide at 25°C, the density of the resulting mixture would be closest to:
 - 1.80 g/mL.
 - 2.20 g/mL.
 - 2.50 g/mL.
 - 2.80 g/mL.



10. According to Table 1, the density of the mixtures increased as the:
- F. mass of the mixtures increased.
 - G. mass of the mixtures decreased.
 - H. volume of the mixtures increased.
 - J. volume of the mixtures decreased.
11. Based on the data in Figure 1, 25 g of powdered sucrose would most likely have a volume closest to:
- A. 8 mL.
 - B. 10 mL.
 - C. 13 mL.
 - D. 15 mL.

Passage III

A pesticide applied one time to orchards located near Site 1 (see Figure 1) at the end of January was detected shortly afterward in the waters of an adjacent river system and bay. Figure 2 shows the data collected daily in February at Site 1: the pesticide concentration in the river water, in nanograms per liter (ng/L); the inches of rain that fell in the area during several storms; and the river discharge (in ft³/sec). *Discharge* is the volume of water passing a given point in a given amount of time. Figure 3 shows pesticide concentrations for February 4 through February 25 at the 3 sites shown in Figure 1.

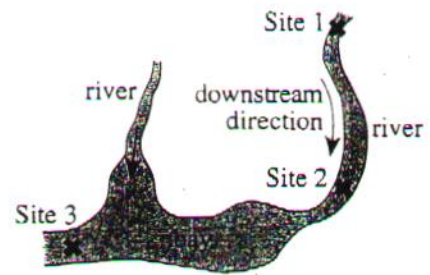


Figure 1

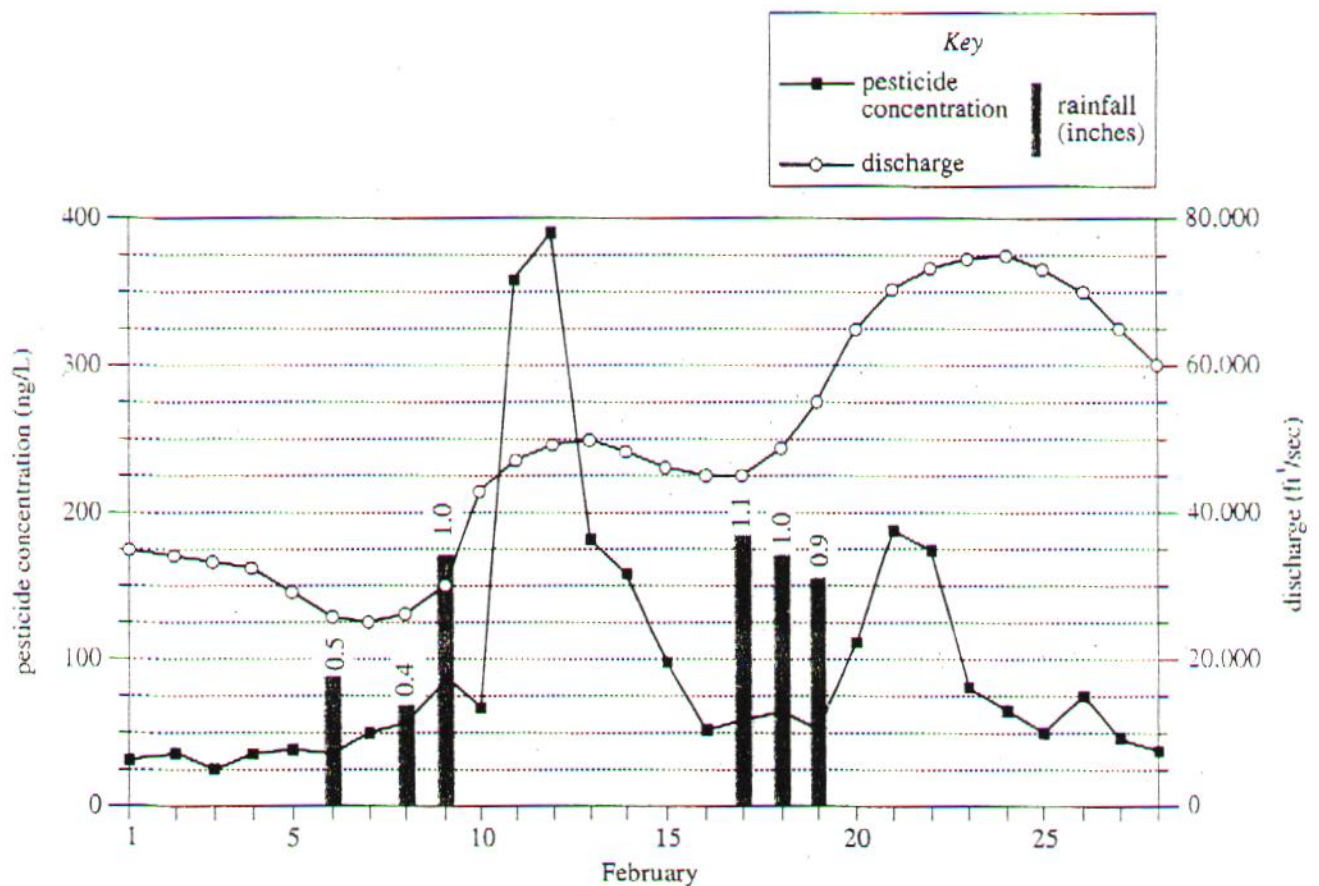


Figure 2

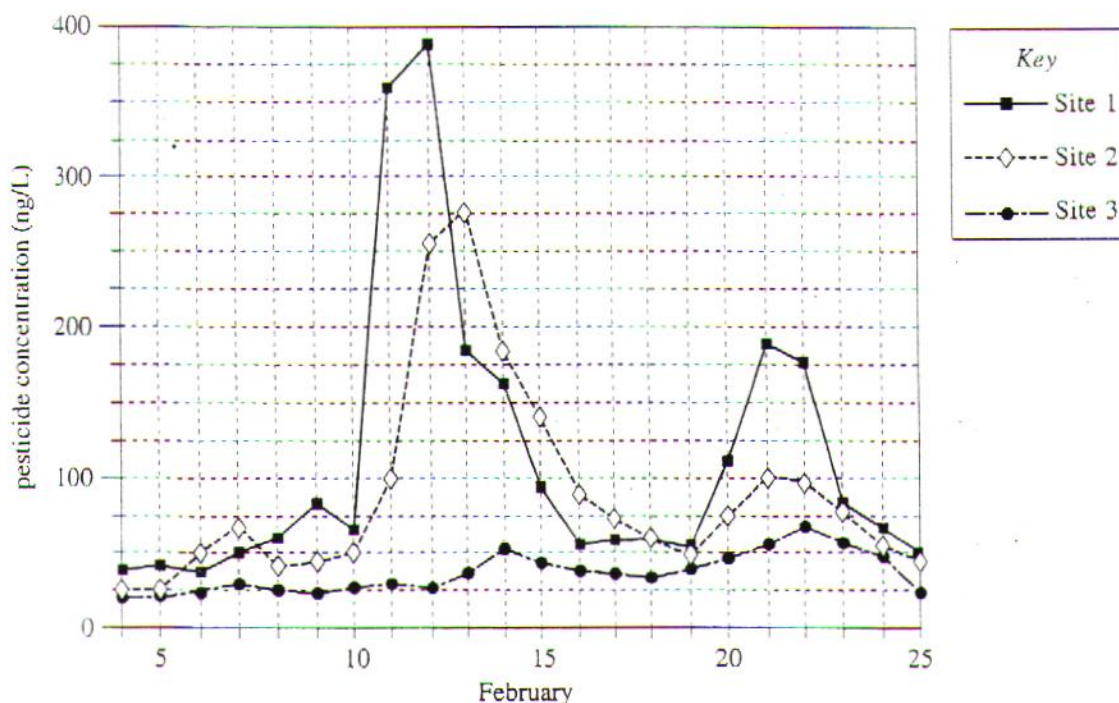


Figure 3

Figures adapted from Kathryn Kuivila, "Diazinon Concentrations in the Sacramento and San Joaquin Rivers and San Francisco Bay, California, January 1993." ©1993 by the U.S. Geological Survey.

12. Which of the following statements best explains why the highest pesticide concentrations for Site 1 and the highest pesticide concentrations for Site 2 shown in Figure 3 were not recorded on the same day?
- The discharge in the river at Site 2 was highest on February 11.
 - The discharge in the river at Site 2 was lowest on February 11.
 - The river water containing the highest pesticide concentration for Site 1 took 1 day to travel downstream to Site 2.
 - The river water containing the highest pesticide concentration for Site 1 took several weeks to travel upstream to Site 2.
13. According to Figure 3, compared to the pesticide concentration measured on a given day at Site 2, the pesticide concentration measured on that same day at Site 1 was:
- always lower.
 - always the same.
 - always higher.
 - sometimes higher and sometimes lower.
14. According to Figure 2, over the first 12 days of February, the discharge in the river:
- increased only.
 - decreased only.
 - increased, then decreased.
 - decreased, then increased.
15. Which of the following figures best represents the pesticide concentrations measured at the 3 sites on February 13?
- -
 -
 -
16. According to Figure 2, the total measured rainfall at Site 1 in February was most likely closest to which of the following?
- 2 inches
 - 3 inches
 - 4 inches
 - 5 inches

Passage IV

A student studied the properties of spring systems using weights and 3 identical, low-mass springs.

Study 1

The student suspended 1 spring from a support and measured the length of the spring. He attached a 10 N (newton) weight to the spring and again measured the length of the spring (see Figure 1).

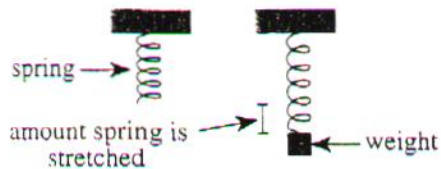


Figure 1

The student calculated the amount the spring was stretched by the weight. He repeated the measurement using other weights. The results are shown in Table 1.

Table 1	
Weight (N)	Amount of stretch (m)
10	0.1
20	0.2
30	0.3

The *spring constant*, K , is the amount of force that must be exerted to stretch a spring a given amount. Based on Table 1, K for this spring was 100 N/m.

Study 2

The student suspended 2 of the springs *in series* (end to end) from the support and measured their amount of stretch with various weights attached. He repeated the procedure with 3 springs in series (see Figure 2).

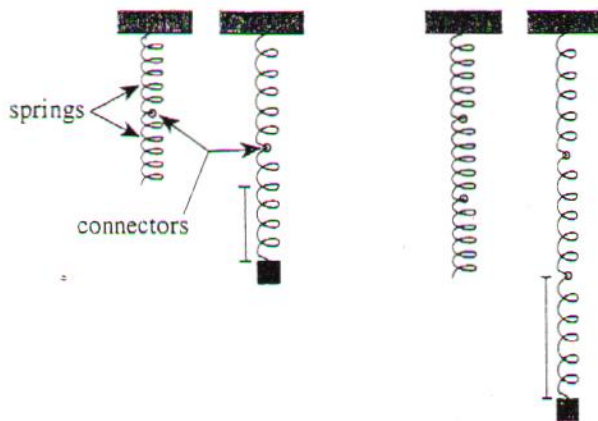


Figure 2

The results are shown in Table 2.

Table 2		
Weight (N)	Amount of stretch (m) with:	
	2 springs in series	3 springs in series
10	0.2	0.3
20	0.4	0.6
30	0.6	0.9

For 2 springs in series, K was 50 N/m; for 3 springs in series, K was 33 N/m.

Study 3

The student suspended 2 of the springs *in parallel* (see Figure 3).

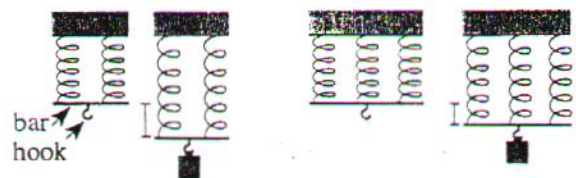


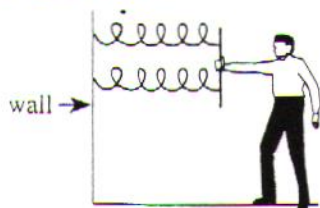
Figure 3

The student measured the amount of stretch with various weights attached. He repeated the procedure with 3 springs in parallel. The results are shown in Table 3.

Table 3		
Weight (N)	Amount of stretch (m) with:	
	2 springs in parallel	3 springs in parallel
10	0.05	0.03
20	0.10	0.07
30	0.15	0.10

For 2 springs in parallel, K was 200 N/m; for 3 springs in parallel, K was 300 N/m.

17. The student connected 2 of the springs in parallel to each other and attached them to hooks screwed into the wall. The student pulled horizontally on the springs, as shown in the figure below.



Based on Study 3, to stretch each spring 0.15 m, the student had to exert how much force?

- A. 5 N
 - B. 10 N
 - C. 20 N
 - D. 30 N
18. The greater the *stiffness* of a spring, the greater the amount of force required to stretch or compress the spring a given amount. Based on the results of the 3 studies, which of the spring systems had the greatest stiffness?
- F. 2 springs in parallel
 - G. 2 springs in series
 - H. 3 springs in parallel
 - J. 3 springs in series
19. Consider 3 springs, identical to the ones used in Studies 1–3, that are connected in parallel. If a 15 N weight had been hung from the springs as in Study 3, how much would any one of the springs have been stretched?
- A. Less than 0.03 m
 - B. Between 0.03 m and 0.07 m
 - C. Between 0.07 m and 0.10 m
 - D. More than 0.10 m

20. Suppose 4 springs identical to the one used in Study 1 are connected in series with each other and suspended from a support as in Study 2. If a 20 N weight is attached to the free end of the springs, the 4 springs will be stretched by a total of:

- F. 0.4 m.
- G. 0.8 m.
- H. 1.2 m.
- J. 2.0 m.

21. Based on the results of Study 1, the spring constant K is calculated by:

- A. adding the weight to the amount of stretch.
- B. subtracting the weight from the amount of stretch.
- C. dividing the weight by the amount of stretch.
- D. multiplying the weight by the amount of stretch.

22. Consider identical springs, each with $K = 200 \text{ N/m}$. Based on Studies 1–3, which of the following combinations of the springs would give a K of 400 N/m?

- F. Two of the springs connected in parallel with each other
- G. Two of the springs connected in series with each other
- H. Three of the springs connected in parallel with each other
- J. Three of the springs connected in series with each other

Passage V

A block is placed on an inclined plane, as shown in Figure 1.

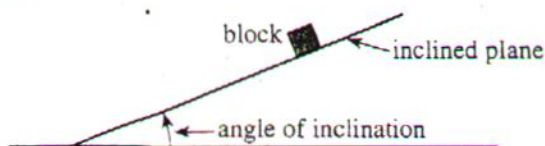


Figure 1

The angle of inclination is increased. The angle at which static friction is overcome and the block starts to slide is α_c , the *critical angle for sliding*. The amount of static friction to be overcome depends on μ , the *coefficient of static friction*.

Table 1 gives α_c and μ for a steel block sliding on various uncoated surfaces at 20°C. Tables 2 and 3 give α_c and μ for the steel block sliding on a steel surface coated with various oils at 20°C and 100°C, respectively.

Trial	Surface composition of inclined plane	α_c (degrees)	μ (20°C)
1	steel	30.1	0.58
2	cast iron	21.8	0.40
3	base white metal	28.8	0.55
4	white metal (tin base)	38.7	0.80

Trial	Steel surface coated with:	α_c (degrees)	μ (20°C)
5	light machine oil	9.1	0.16
6	thick gear oil	7.1	0.13
7	heavy motor oil	11.3	0.20
8	B.P. paraffin oil	10.2	0.18

Trial	Steel surface coated with:	α_c (degrees)	μ (100°C)
9	light machine oil	10.8	0.19
10	thick gear oil	8.5	0.15
11	heavy motor oil	11.9	0.21
12	B.P. paraffin oil	12.4	0.22

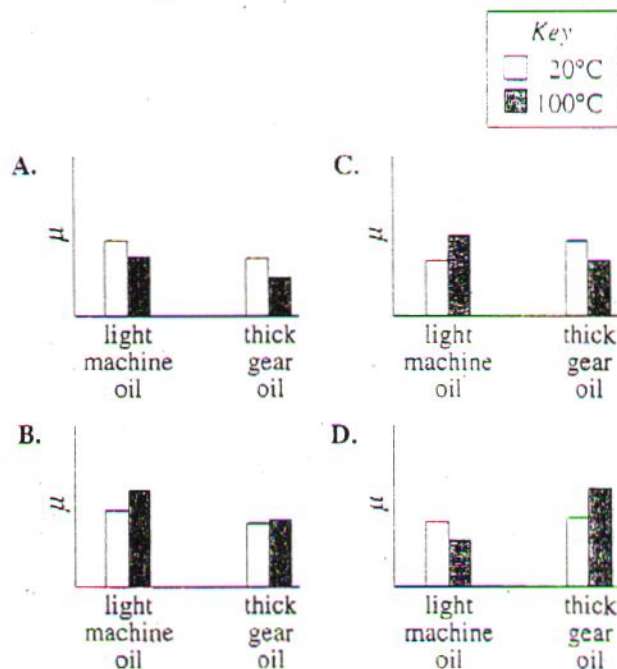
23. Raising the temperature of the oil coating the steel surface from 20°C to 100°C resulted in the greatest amount of change in μ for:

A. light machine oil.
 B. thick gear oil.
 C. heavy motor oil.
 D. B.P. paraffin oil.

24. Based on the passage and the data in Tables 1-3, to start the block sliding down the inclined plane, the greatest amount of force was required in:

F. Trial 1.
 G. Trial 4.
 H. Trial 6.
 J. Trial 9.

25. Based on the data in Tables 2 and 3, which of the following bar graphs best represents μ for light machine oil and thick gear oil at 20°C and 100°C?



26. Based on the results of Trials 1–8, for a steel block placed on a cast iron surface coated with light machine oil, μ would be:

- F. less than 0.40.
- G. between 0.40 and 0.55.
- H. between 0.55 and 0.80.
- J. greater than 0.80.

27. Do the data in Tables 1–3 support the hypothesis that for the block on a given surface, μ varies with the mass of the block?

- A. Yes, because the mass of the block was varied in Trials 1–12.
- B. Yes, because the mass of the block was not varied in Trials 1–12.
- C. No, because the mass of the block was varied in Trials 1–12.
- D. No, because the mass of the block was not varied in Trials 1–12.

Passage VI

Eating chips of lead-based paint is a cause of lead poisoning in children. According to the law, paint is considered lead-based if the lead content exceeds 0.5% by weight. *Atomic-absorption spectroscopy* (AAS) is used to determine the lead content of paint. Samples containing lead in solution are vaporized in a flame, and light of a single wavelength characteristic of lead is shone through the flame. The *absorbance* measures the amount of light absorbed by the lead in the flame. The following experiments were done using AAS.

Table 1			
Location	House	Absorbance	% lead
Interior walls	A	0.404	5.67
	B	0.000	0.00
Exterior walls	A	0.596	8.36
	B	0.001	0.02
Outdoor furniture	A	0.842	11.80
	B	0.029	0.41

Experiment 1

Solutions with known concentrations of lead were prepared and their absorbances were measured (see Figure 1).

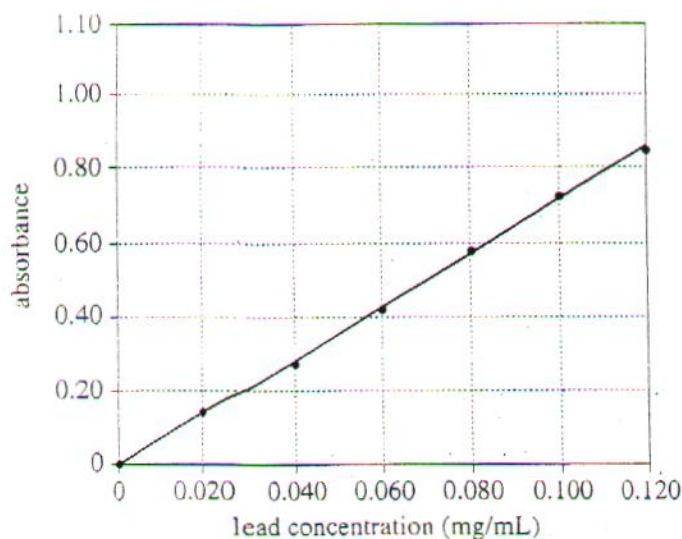


Figure 1

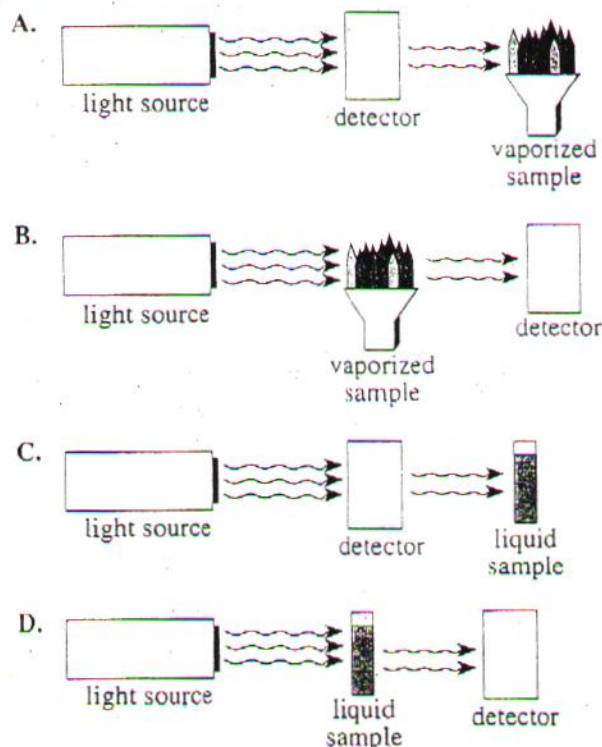
Experiment 2

Paint chips were collected from interior walls and exterior walls and from outdoor furniture at House A, built in 1945. The paint chips from a given location were powdered. Then 100 mg samples of the powder were treated with nitric acid and heated for at least 30 minutes to convert all of the lead to soluble lead nitrate. The solutions were evaporated to a dry powder, then dissolved in hydrochloric acid and diluted to 100 mL with H_2O . The absorbances were measured and the lead concentrations were determined using the data in Figure 1. The % lead (average weight percent of lead in the paint) at the location was then calculated. The same analysis was performed at House B, built in 1990 (see Table 1).

28. The main purpose of Experiment 1 was to establish:

- F. the relationship between lead concentration and absorbance.
- G. if AAS could be used to determine lead concentrations of solutions of paint chips.
- H. how paint chips taken from different locations vary in lead concentration.
- J. that solutions with known lead concentrations could be prepared.

29. Which of the following diagrams best shows how the solutions were analyzed using the AAS instrument in Experiments 1 and 2?



30. If, in Experiment 1, an additional solution with a lead concentration of 0.14 mg/mL had been analyzed by AAS, the absorbance of this solution would most likely have been closest to:
- F. 0.80.
 - G. 0.90.
 - H. 1.00.
 - J. 1.10.
31. Paint chips from a storage shed at House A were analyzed as in Experiment 2. The % lead was found to be higher than was measured for the exterior walls of House A, but lower than was measured for the outdoor furniture. Which of the following values could represent the average absorbance reading for the paint chips from the storage shed?
- A. 0.50
 - B. 0.70
 - C. 0.90
 - D. 1.00
32. House C was built in 1971. Which of the following conclusions about the % lead of paint chips collected from the interior walls at House C is most valid based on the results of Experiment 2?
- F. The % lead would be lower than that calculated for the interior walls at both Houses A and B.
 - G. The % lead would be higher than that calculated for the interior walls at both Houses A and B.
 - H. The % lead would be greater than that calculated for the interior walls at House A, but lower than that calculated for the interior walls at House B.
 - J. The % lead cannot be reliably predicted using the data collected for the interior walls at Houses A and B.
33. According to the results of Experiment 1, in the AAS instrument, as the lead concentration increased for each sample, the amount of light passing through the flame that was measured for each sample:
- A. increased only.
 - B. decreased only.
 - C. increased, then decreased.
 - D. decreased, then increased.

Passage VII

PTC is a chemical found in some foods. However, not all humans can taste PTC. The ability to taste PTC is determined by 1 gene (Gene P). Gene P has 2 alleles (possible DNA sequences), which are designated P and p.

Four students disagree on 2 questions about Gene P.

- Which alleles are carried by PTC tasters and by PTC nontasters?
- Is Gene P X-linked (on the X chromosome), or is it autosomal (not found on a sex chromosome)?

The students know that if Gene P is X-linked, females have 2 copies of it and males have 1. (Females receive 1 X chromosome from their mother and 1 X chromosome from their father. Males receive 1 X chromosome from their mother, but no X chromosomes from their father.) They also know that if Gene P is autosomal, each person has 2 copies of it.

Student 1

Gene P is autosomal. Tasters have 1 or 2 P alleles. The P allele codes for (leads to the production of) a protein. This protein is an enzyme required for producing PTC receptors. Without these receptors, one cannot taste PTC. The p allele codes for a different protein that cannot act as a PTC receptor.

Student 2

Gene P is autosomal. The P allele codes for a protein that inhibits the biochemical pathway leading to the production of PTC receptors. The p allele codes for a protein that does not inhibit the production of PTC receptors. Individuals with either 1 or 2 P alleles do not produce PTC receptors and do not taste PTC. In individuals with no P alleles, the pathway is not interrupted, PTC receptors are produced, and PTC can be tasted.

Student 3

Gene P is X-linked. The P allele codes directly for the PTC receptor. So, tasters must have at least 1 P allele and nontasters lack P alleles. For tasters, sensitivity to PTC varies. Tasters with 2 P alleles produce more PTC receptors and can taste PTC at lower concentrations than can tasters with 1 P allele.

Student 4

Gene P is X-linked. However, it does not affect the production of PTC receptors. The P allele codes for a protein that binds to PTC receptors. When this protein is present, PTC cannot bind to PTC receptors and PTC cannot be tasted. The p allele codes for a different protein that does not bind to PTC receptors.

34. Student 1 and Student 3 would agree that to produce PTC receptors an individual must have at least:

F. 1 p allele.
G. 1 P allele.
H. 2 p alleles.
J. 2 P alleles.

35. According to Student 2, which of the following tables correctly shows the relationship between the alleles present at Gene P and the ability to taste PTC?

A.

Gene P alleles	Tasters or nontasters
P and P	nontasters
P and p	nontasters
p and p	tasters

B.

Gene P alleles	Tasters or nontasters
P and P	nontasters
P and p	tasters
p and p	tasters

C.

Gene P alleles	Tasters or nontasters
P and P	tasters
P and p	nontasters
p and p	nontasters

D.

Gene P alleles	Tasters or nontasters
P and P	tasters
P and p	tasters
p and p	nontasters

36. According to the hypothesis presented by Student 1, which of the following individuals CANNOT taste PTC?

F. A man with 2 p alleles
G. A man with 1 P allele and 1 p allele
H. A woman with 2 P alleles
J. A woman with 1 P allele and 1 p allele

37. Which 2 students would most likely argue that nontasters can become tasters by blocking the production of the protein coded for by the P allele?

A. Student 1 and Student 2
B. Student 1 and Student 3
C. Student 2 and Student 4
D. Student 3 and Student 4



38. Based on the argument presented by Student 3, which of the following individuals would most likely be able to taste the *lowest* concentration of PTC?
- F. A man with no p alleles
 - G. A man with no P alleles
 - H. A woman with no p alleles
 - J. A woman with no P alleles
39. Suppose additional evidence indicated that tasters and nontasters have, on average, about the same number of PTC receptors. This finding would best support the argument presented by:
- A. Student 1.
 - B. Student 2.
 - C. Student 3.
 - D. Student 4.
40. A male taster and a female nontaster produced 4 girls and 4 boys. If Student 3 is correct, which of these children are tasters and which are nontasters?
- F. The 4 girls are tasters and the 4 boys are nontasters.
 - G. The 4 girls are nontasters and the 4 boys are tasters.
 - H. All of the children are tasters.
 - J. All of the children are nontasters.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Student Review Questions

The test supervisor will give you instructions for responding to these questions at the end of the test session.

Questions 1–6 ask about your experience at the test center today.

1. Did the reporting location printed on your admission ticket direct you to the proper place to check in?
2. Were you either recognized by the testing staff or asked for identification?
3. Were you directed to a specific seat in the test room?
4. Were you seated at least three feet from the nearest examinee?
5. Did you have enough writing space on your desk or table to mark your answer document?
6. Were all the examinees in your room seated facing the same direction?

Questions 7–9 ask about your use of calculators.

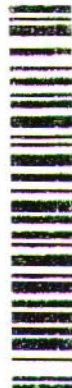
7. Did you bring a calculator to use on today's test?
8. Did you bring a calculator with trigonometric functions (for example, cos), parentheses, or exponent capabilities to use on today's test?
9. Did you use a calculator today to help you on one or more questions on the ACT Mathematics Test?

Questions 10–11 ask about your test preparation activities.

10. Did you use test preparation software or materials available directly from ACT (for example, ACT Online Prep™ or *The Real ACT Prep Guide*) to help you get ready for taking the ACT tests?
11. Did you use any test preparation software or materials—other than those available directly from ACT—to help you get ready for taking the ACT tests?

Questions 12–15 ask about the education of your parents/guardians.

12. Did your Mother/Guardian 1 graduate from high school or earn a GED?
 13. Did your Mother/Guardian 1 graduate from a four-year college?
 14. Did your Father/Guardian 2 graduate from high school or earn a GED?
 15. Did your Father/Guardian 2 graduate from a four-year college?
-



Your Signature (do not print): _____

Print Your Name Here: _____

Your Date of Birth:		
10	15	1996
Month	Day	Year

Form 11S

The ACT® Writing Test Booklet

You must take the multiple-choice tests before you take the Writing Test.

Directions

This is a test of your writing skills. You will have thirty (30) minutes to write an essay in English. Before you begin planning and writing your essay, read the writing prompt carefully to understand exactly what you are being asked to do. Your essay will be evaluated on the evidence it provides of your ability to express judgments by taking a position on the issue in the writing prompt; to maintain a focus on the topic throughout the essay; to develop a position by using logical reasoning and by supporting your ideas; to organize ideas in a logical way; and to use language clearly and effectively according to the conventions of standard written English.

You may use the unlined pages in this test booklet to plan your essay. These pages will not be scored. *You must write your essay in pencil on the lined pages in the answer folder.* Your writing on those lined pages will be scored. You may not need all the lined pages, but to ensure you have enough room to finish, do NOT skip lines. You may write corrections or additions neatly between the lines of your essay, but do NOT write in the margins of the lined pages. *Illegible essays cannot be scored, so you must write (or print) clearly.*

If you finish before time is called, you may review your work. Lay your pencil down immediately when time is called.

DO NOT OPEN THIS BOOKLET UNTIL TOLD TO DO SO.

ACT Writing Test Prompt

Many large school districts allow students to attend magnet high schools that offer specialized academic programs in areas such as mathematics or performing arts. Some educators support magnet high schools because they think it is beneficial to allow high school students to pursue special academic interests or talents. Other educators do not support magnet high schools because they think it is not beneficial to separate students according to their academic interests and talents. In your opinion, should high school students be allowed to attend magnet schools?

In your essay, take a position on this question. You may write about either one of the two points of view given, or you may present a different point of view on this question. Use specific reasons and examples to support your position.

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✧ **ACT English Practice Tests:**

<https://www.crackab.com/act/english/>

✧ **ACT Math Practice Tests:**

<https://www.crackab.com/act/math/>

✧ **ACT Reading Practice Tests:**

<https://www.crackab.com/act/reading/>

✧ **ACT Science Practice Tests:**

<https://www.crackab.com/act/science/>

ACT Grammar: <https://www.crackab.com/act/grammar/>

ACT Real Past Papers Download:

<https://www.crackab.com/act-downloads/>

Digital SAT & New SAT Practice Tests:

<https://www.cracksat.net>

Real SAT Tests Download:

<http://www.cracksat.net/sat-downloads/>

AP Exams Practice Tests:

<https://www.crackap.com>

<https://www.apstudy.net>

OPEN **ACT** provides you the answers for ACT Form 62D:

<u>English</u>		<u>Math</u>	<u>Reading</u>	<u>Science</u>
1) C	45) A	Missing	1) –	1) A
2) G	46) H		2) –	2) J
3) B	47) B		3) –	3) B
4) G	48) G		4) –	4) F
5) D	49) D		5) –	5) B
6) J	50) F		6) –	6) H
7) B	51) A		7) –	7) C
8) J	52) J		8) –	8) G
9) B	53) D		9) –	9) C
10) F	54) G		10) –	10) J
11) C	55) C		11) D	11) D
12) G	56) F		12) F	12) H
13) D	57) A		13) C	13) D
14) J	58) F		14) J	14) J
15) B	59) B		15) D	15) A
16) G	60) J		16) F	16) J
17) A	61) B		17) B	17) D
18) H	62) H		18) G	18) H
19) D	63) D		19) B	19) B
20) H	64) F		20) F	20) G
21) D	65) A		21) A	21) C
22) F	66) F		22) F	22) F
23) B	67) C		23) C	23) D
24) J	68) H		24) J	24) G
25) C	69) A		25) B	25) B
26) G	70) F		26) H	26) F
27) C	71) B		27) C	27) D
28) J	72) H		28) J	28) F
29) D	73) C		29) A	29) B
30) H	74) J		30) G	30) H
31) A	75) D		31) A	31) B
32) J			32) J	32) J
33) C			33) A	33) B
34) F			34) G	34) –
35) A			35) B	35) –
36) J			36) H	36) –
37) C			37) C	37) –
38) F			38) G	38) –
39) D			39) A	39) –
40) J			40) J	40) –
41) C				
42) G				
43) B				
44) J				